

# SUBMITTAL RESPONSE

Norcross, GA — Lidl #1517 | Job 15170000GR  
4990 Jimmy Carter Blvd, Norcross, GA 30093



|                 |                                                                                                                                                                 |              |                                                     |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------------------------------------------|
| To              | Coles Construction — Clay Hotchkiss,<br>Preconstruction Manager (470-495-7547)                                                                                  | Date         | October 5, 2026                                     |
| From            | Place Services, Inc. — Luis Marcelino                                                                                                                           | CC           | Bobby McTeague, Tucker Teague, Tim<br>Mauldin (PSI) |
| PSI Log #       | SUB-002 (Rev 0)                                                                                                                                                 | Procore #    | 33 4100-2.0                                         |
| Fieldwire #     | PSI-0002 — Stormwater Package                                                                                                                                   | Spec Section | 33 41 00 — Storm Drainage                           |
| Description     | Hydrodynamic Separator (First Defense, ST8) / Water Quality Filter (Up-Flo, ST5) / Underground Detention System (CMP) — Hydro International / Southeast Culvert |              |                                                     |
| Submitted       | 8/26/2026 (Coles package dated<br>8/26/2026)                                                                                                                    | Engineer     | CHA Consulting — Chris Manes; reviewed<br>10/1/2026 |
| Engineer Action | <b>REJECTED — resubmission required</b>                                                                                                                         | GC Action    | <b>REJECTED — Revise &amp; Resubmit</b>             |

## Engineer's Comment (CHA review form, 10/1/2026)

"Items in this submittal will need to be updated based on the revised site plan layout the is pending County review."

## Item-by-Item Response

| # | Item / Structure                                                                                                 | Pkg<br>pages | Engineer markup                                                                                                                        | Required action — Coles                                                                                                                                      |
|---|------------------------------------------------------------------------------------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Hydrodynamic Separator —<br>Hydro International First<br>Defense, 5-ft dia., Structure<br>ST8 (dwg FD GA-5)      | 2–21         | Sheet crossed out — "Feature<br>is no longer required."                                                                                | <b>Remove ST8 from the stormwater scope. Do<br/>not order or fabricate. Omit from the<br/>resubmittal.</b>                                                   |
| 2 | Water Quality Filter — Hydro<br>International Up-Flo Filter,<br>96x96 precast vault, 6<br>modules, Structure ST5 | 22–47        | Sheet crossed out — "Feature<br>to be updated based on revised<br>civil design plans that will be<br>issued at a later date."          | <b>Hold — do not release for fabrication. Re-size<br/>and resubmit per the revised civil plans (sizing<br/>calcs, vault dwg, rim/inverts, module count).</b> |
| 3 | Underground Detention<br>System — Southeast Culvert<br>84" CMP, 14 ga, aluminized<br>Type 2 (sheets 1–12)        | 48–59        | Cover sheet crossed out —<br>"Feature to be updated based<br>on revised civil design plans<br>that will be issued at a later<br>date." | <b>Hold — do not release for fabrication. Revise<br/>layout, lengths, risers and manifolds per the<br/>revised civil plans and resubmit.</b>                 |

## Resubmittal Requirements

- Do not order, fabricate or release any item in this package. The design is being revised, and the revised site plan is pending Gwinnett County review.
- PSI will send Coles the revised civil design plans when they are issued. Resubmit against those plans; the submittal dated 8/26/2026 is not to be used.
- Resubmit as **Rev 1** under the same submittal number (PSI-0002 / 33 4100-2.0). Include current sizing calculations, structure drawings showing rims, inverts, pipe sizes and orientation, and the CMP layout. Cloud all changes from Rev 0.
- Confirm lead times for the revised Up-Flo vault and CMP detention system when you resubmit.

## Enclosures

CHA Submittal / Shop Drawing Processing Form (10/1/2026) and CHA-marked submittal package, 61 pages, merged behind this sheet (file: SUB-002\_R0\_CHA\_Reviewed\_REJECTED\_2026-10-01.pdf).

## Place Services, Inc.

By: \_\_\_\_\_  
Luis Marcelino

Date: \_\_\_\_\_

PSI review does not relieve the subcontractor of responsibility for conformance with the Contract Documents, field verification of dimensions, or coordination with other trades.



# Submittal/ Shop Drawing Processing Form

Single-Discipline Review

|                                                                                                                 |                    |                         |                                                                     |                         |
|-----------------------------------------------------------------------------------------------------------------|--------------------|-------------------------|---------------------------------------------------------------------|-------------------------|
| DATE SENT BY CONTRACTOR:                                                                                        | 8/26/26            | PROJECT NAME:           | Norcross Lidl                                                       | PAGE 1 OF 61            |
| ID NUMBER:                                                                                                      | PSI-0002           | IS THIS A RESUBMISSION? | <input type="checkbox"/> YES <input checked="" type="checkbox"/> NO | CHA PROJECT No.: 041386 |
| DESCRIPTION:                                                                                                    | Stormwater Package |                         |                                                                     |                         |
| (PROVIDE INFORMATION ABOVE AS IT APPEARS ON THE SUBMITTAL/SHOP DRAWING TRANSMITTAL AND ATTACH THIS FORM TO IT.) |                    |                         |                                                                     |                         |

|                                                                                                                                                                                                                                        |     |           |     |       |           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----------|-----|-------|-----------|
| TO CHA BUSINESS PRACTICE:                                                                                                                                                                                                              | LDV | REVIEWER: | CLM | DATE: | 10/1/2026 |
| <b>REVIEWER'S COMMENTS:</b><br>Items in this submittal will need to be updated based on the revised site plan layout the is pending County review.                                                                                     |     |           |     |       |           |
| DOES THIS SUBMITTAL/ SHOP DRAWING HAVE TO BE REVIEWED BY ANYONE ELSE? <input checked="" type="checkbox"/> No <input type="checkbox"/> YES <b>WHO?</b> _____                                                                            |     |           |     |       |           |
| FOR CHA USE ONLY REVIEW STATUS (CHECK ONE): <input type="checkbox"/> NET <input type="checkbox"/> MCN <input checked="" type="checkbox"/> REJ* <input type="checkbox"/> SSI* <input type="checkbox"/> R&R* <input type="checkbox"/> NR |     |           |     |       |           |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| <b>REVIEW STATUS:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                           |
| <input type="checkbox"/> NET = NO EXCEPTIONS TAKEN (APPROVED)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <input type="checkbox"/> MCN = MAKE CORRECTIONS NOTED (APPROVED AS NOTED) |
| <input type="checkbox"/> REJ* = REJECTED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <input type="checkbox"/> SSI* = SUBMIT SPECIFIED ITEM                     |
| <input type="checkbox"/> R&R* = REVISE AND RESUBMIT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <input type="checkbox"/> NR = NOT REVIEWED/INFORMATION ONLY               |
| <b>*RESUBMISSION REQUIRED IF REVIEW STATUS IS R&amp;R, REJ, OR SSI</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                           |
| THIS SUBMITTAL/ SHOP DRAWING HAS BEEN REVIEWED FOR GENERAL CONFORMANCE WITH THE DESIGN CONCEPT OF THE PROJECT AND GENERAL COMPLIANCE WITH THE INFORMATION PROVIDED IN THE CONTRACT DOCUMENTS. ANY ACTION SHOWN IS SUBJECT TO THE REQUIREMENTS OF THE PLANS AND SPECIFICATIONS. CONTRACTOR (OR DESIGNEE) IS RESPONSIBLE FOR THE FOLLOWING: DIMENSIONS WHICH SHALL BE CONFIRMED AND CORRELATED AT THE JOB SITE, FABRICATION PROCESSES AND TECHNIQUES OF CONSTRUCTION, COORDINATION OF WORK WITH THAT OF ALL OTHER TRADES, AND THE SATISFACTORY PERFORMANCE OF THE WORK. CONTRACTOR (OR DESIGNEE) IS NOT RELIEVED OF RESPONSIBILITY FOR CONFORMANCE WITH DESIGN DRAWINGS, SPECIFICATIONS, AND APPLICABLE CODES, ALL OF WHICH HAVE PRIORITY OVER THIS SUBMITTAL/ SHOP DRAWING. "NO EXCEPTION TAKEN" OR "MAKE CORRECTIONS NOTED" SHALL NOT RELIEVE THE CONTRACTOR (OR DESIGNEE) FROM RESPONSIBILITY FOR OMISSIONS, OR ERRORS IN DIMENSIONS, SHOP FITS, FIELD CONNECTIONS, ETC.; OR FOR PROVIDING THE PROPER QUANTITY OF MATERIALS; OR FOR COMPLIANCE WITH THE CONTRACT; OR FOR THE SUCCESSFUL COMPLETION OF THE PROJECT. |                                                                           |
| REVIEWED BY:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Chris Manes                                                               |
| DATE:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 10/1/2026                                                                 |

Lidl 1517 Norcross, GA - Ground Up

4990 Jimmy Carter Blvd. Norcorss, GA 30093

Report generated by Bobby McTeague (Place Services, Inc.)



Submittal #PSI-0002 - Stormwater Package

|             |                                       |                  |                         |
|-------------|---------------------------------------|------------------|-------------------------|
| Creator     | Bobby McTeague (Place Services, Inc.) | Submission due   | -                       |
| Assignee    | Bobby McTeague (Place Services, Inc.) | Response due     | 09/09/2026              |
| Reviewers   | -                                     | Closed on        | -                       |
| Status      | Pending approval                      | Reference number | -                       |
| Type        | Product data                          | Specification    | 33 41 00 STORM DRAINAGE |
| Description | -                                     |                  |                         |

Approval (rev. 0)

|              |   |              |   |
|--------------|---|--------------|---|
| Responded by | - | Responded on | - |
| Response     | - |              |   |
| Notes        | - |              |   |
| Attachments  | - |              |   |

Submission (rev. 0)

|              |                                                                                                                                                                  |              |            |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------|
| Submitted by | Bobby McTeague (Place Services, Inc.)                                                                                                                            | Submitted on | 08/26/2026 |
| Notes        | -                                                                                                                                                                |              |            |
| Attachments  | PDF attachments are included in this export.<br><a href="#">Lidl-1517-Norcross-GA-Ground-Up - Submittal-PSI-0002-Stormwater-Package-08-26-26.pdf</a>   (Page 2 ) |              |            |

# SUBMITTAL PACKAGE

## Lidl Grocery Store #1517

4990 Jimmy Carter Blvd, Norcross, GA 30093 — Gwinnett County

|                      |                                                                            |
|----------------------|----------------------------------------------------------------------------|
| <b>Prepared By:</b>  | Coles Construction                                                         |
| <b>Contact:</b>      | Clay Hotchkiss, Preconstruction Manager   470-495-7547                     |
| <b>Submitted To:</b> | GC Grocery Construction (Place Services Inc.)                              |
| <b>Attn:</b>         | Luis Marcelino / Tucker Teague                                             |
| <b>Scope:</b>        | Stormwater Management Submittals - Hydro International / Southeast Culvert |

### Table of Contents

| Section                                                   | Pages |
|-----------------------------------------------------------|-------|
| 1. Hydrodynamic Separator (First Defense) - Structure ST8 | 2–21  |
| 2. Water Quality Filter (Up-Flo Filter) - Structure ST5   | 22–47 |
| 3. Underground Detention System (CMP)                     | 48–59 |

#### SHOP DRAWING / SUBMITTAL REVIEW

- ☒ NO EXCEPTION TAKEN  
☐ REVISE AND RESUBMIT  
☐ MAKE CORRECTIONS NOTED  
☐ REJECTED

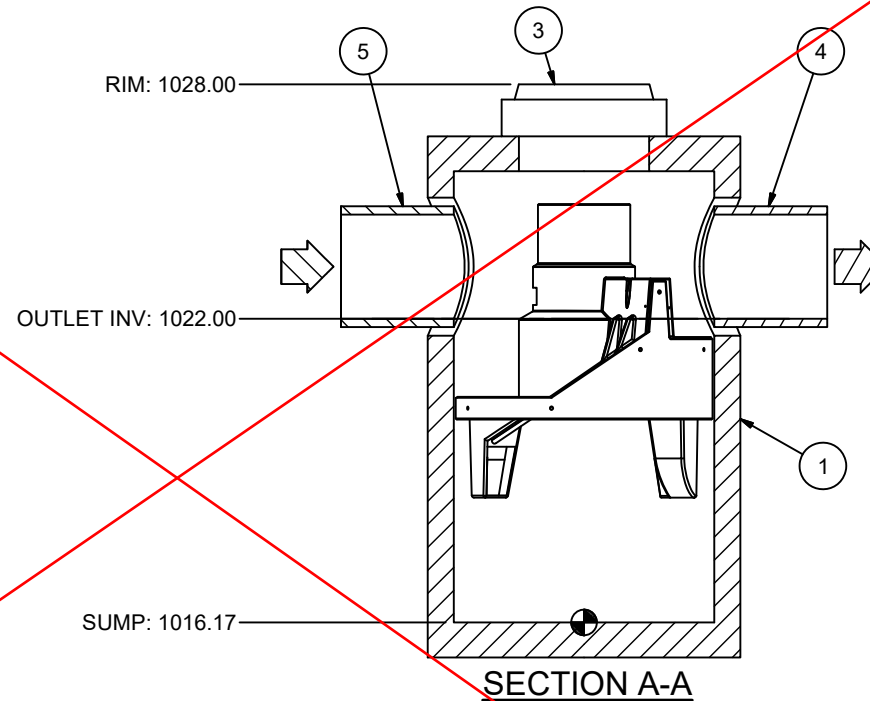
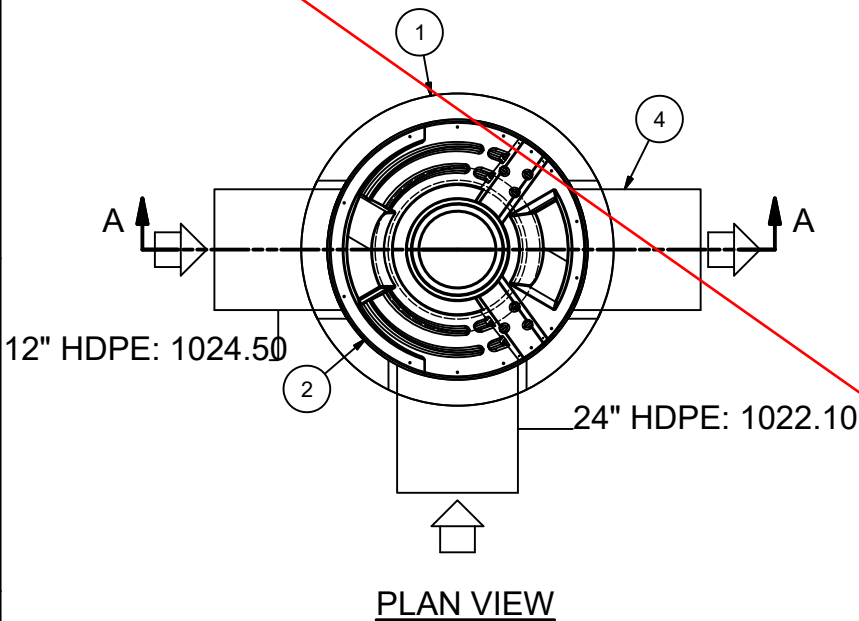
SUBMITTAL WAS REVIEWED FOR DESIGN CONFORMITY AND GENERAL CONFORMANCE TO CONTRACT DOCUMENTS ONLY. THE CONTRACTOR IS RESPONSABLE FOR CONFIRMING AND CORRELATING DIMENSION AT JOBSITE FOR TOLERANCE, CLEARANCE, QUANTITIES, FABRICATION PROCESSES AND TECHNIQUES OF CONSTRUCTION, COORDINATION OF HIS WORK WITH OTHER TRADES AND FULL COMPLIANCE WITH CONTRACT DOCUMENTS.

By: Bobby McTeague

Date: 08-26-2026

Place Services, Inc  
Canton, GA 30115

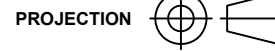
Feature is no longer required.



1. MANHOLE WALL AND SLAB THICKNESSES ARE NOT TO SCALE.

2. CONTACT HYDRO INTERNATIONAL FOR A BOTTOM OF STRUCTURE ELEVATION PRIOR TO SETTING FIRST DEFENSE MANHOLE.

3. CONTRACTOR TO CONFIRM RIM, PIPE INVERTS, PIPE DIA. AND PIPE ORIENTATION PRIOR TO RELEASE OF UNIT TO FABRICATION.



IF IN DOUBT ASK

DATE: 3/17/23 SCALE: 1:30

DRAWN BY: MHM CHECKED BY: MRJ APPROVED BY:

Title  
5-ft DIAMETER  
FIRST DEFENSE  
STR 8

LIDL #1517

NORCROSS, GA

**Hydro**  
International

hydro-int.com

HYDRO INTERNATIONAL

WEIGHT: MATERIAL:

STOCK NUMBER:

DRAWING NO.:  
FD GA-5

SHEET SIZE: B SHEET: 1 OF 1 Rev: -

#### PRODUCT SPECIFICATION:

1. Peak Hydraulic Flow: 20.0 cfs (566 l/s)
2. Min Sediment Storage Capacity: 1.1 cu. yd. (0.8 cu. m.)
3. Maximum Inlet/Outlet Pipe Diameters: 24 in. (600 mm)
4. The treatment system shall use an induced vortex to separate pollutants from stormwater runoff.
5. For more product information including regulatory acceptances, please visit <https://hydro-int.com/en/products/first-defense>

#### GENERAL NOTES:

1. General Arrangement drawings only. Contact Hydro International for site specific drawings.
2. The diameter of the inlet and outlet pipes may be no more than 24".
3. Multiple inlet pipes possible (refer to project plan).
4. Inlet/outlet pipe angle can vary to align with drainage network (refer to project plans).
5. Peak flow rate and minimum height limited by available cover and pipe diameter.
6. Larger sediment storage capacity may be provided with a deeper sump depth.

#### PARTS LIST

| ITEM | QTY | SIZE (in) | SIZE (mm) | DESCRIPTION                         |
|------|-----|-----------|-----------|-------------------------------------|
| 1    | 1   | 60        | 1500      | I.D. PRECAST MANHOLE                |
| 2    | 1   |           |           | INTERNAL COMPONENTS (PRE-INSTALLED) |
| 3    | 1   | 2x2       |           | FRAME/GRATE                         |
| 4    | 1   | 24        |           | OUTLET PIPE (CMP)                   |
| 5    | 1   | 24        |           | INLET PIPE (HDPE)                   |

**Project:** Lidl #1517  
**Location:** Norcross, GA  
**Prepared For:** CHA



**Purpose:** To calculate the Water Quality Peak Flow associated with the Georgia Stormwater Management Manual Requirements. In this situation the WQV to be analyzed is the runoff produced by the first 1.2" of rainfall.

**Reference:** United States Department of Agriculture Natural Resources Conservation Service TR-55 Manual and Georgia Stormwater Management Manual.

**Given:**

| Structure Name | A (acres) | A (miles <sup>2</sup> ) | Runoff Coefficient | Percent Imp. (%)* | t <sub>c</sub> (min) | t <sub>c</sub> (hr) |
|----------------|-----------|-------------------------|--------------------|-------------------|----------------------|---------------------|
| ST8            | 1.68      | 0.00263                 | 0.725              | 75.00             | 5.0                  | 0.083               |
|                |           |                         |                    |                   |                      |                     |
|                |           |                         |                    |                   |                      |                     |
|                |           |                         |                    |                   |                      |                     |
|                |           |                         |                    |                   |                      |                     |

\* Assumes runoff coefficient of 0.95 for impervious surfaces and 0.3 for pervious surfaces.

**Procedure:** The Water Quality Flow (WQF) is calculated using the Water Quality Volume (WQV). This WQV, converted to watershed inches, is substituted for the runoff depth (Q) in the Natural Resources Conservation Service (formerly Soil Conservation Service), TR-55 Graphical Peak Discharge Method.

1. Compute WQV in watershed inches using the following equation:

$$WQV(in) = P * R$$

where: WQV = water quality volume (watershed inches)

P = design precipitation (inches)

R = volumetric runoff coefficient =  $0.05 + 0.009(I)$

I = Impervious cover (Percentage)

| Structure Name | Percent Imp. (%) | R     | P (in) | WQV (in) |
|----------------|------------------|-------|--------|----------|
| ST8            | 75.00            | 0.725 | 1.20   | 0.870    |
|                |                  |       |        |          |
|                |                  |       |        |          |
|                |                  |       |        |          |
|                |                  |       |        |          |

2. Compute the NRCS Runoff Curve Number (CN) using the following equation, or graphically using Figure 2-1 from TR-55 (USDA, 1986):

$$CN = 1000 / [10 + 5P + 10Q - 10(Q^2 + 1.25QP)^{1/2}]$$

where: CN = Runoff Curve Number

P = design precipitation (inches)

Q = runoff depth (watershed inches)

| Structure Name | Q (in) | CN |
|----------------|--------|----|
| ST8            | 0.870  | 97 |
|                |        |    |
|                |        |    |
|                |        |    |
|                |        |    |

3. Using computed CN, read initial abstraction ( $I_a$ ) from Table 4-1 in Chapter 4 of TR-55; compute  $I_a/P$ , interpolating when appropriate.

| Structure Name | $I_a$ (in) | $I_a/P$ |
|----------------|------------|---------|
| ST8            | 0.062      | 0.052   |
|                |            |         |
|                |            |         |
|                |            |         |
|                |            |         |

4. Compute the time of concentration ( $t_c$ ) in hours and the drainage area in square miles. A minimum  $t_c$  of 0.1 hours (6 minutes) should be used.

| Structure Name | $t_c$ (hr) | A (miles <sup>2</sup> ) |
|----------------|------------|-------------------------|
| ST8            | 0.083      | 0.00263                 |
|                |            |                         |
|                |            |                         |
|                |            |                         |
|                |            |                         |

5. Read the unit peak discharge ( $q_u$ ) from Exhibit 4-II or 4-III in Chapter 4 of TR-55 for appropriate  $t_c$  for type II rainfall distribution.

| Structure Name | $t_c$ (hr) | $I_a/P$ | $q_u$ (csm/in) |
|----------------|------------|---------|----------------|
| ST8            | 0.083      | 0.052   | 1000           |
|                |            |         |                |
|                |            |         |                |
|                |            |         |                |
|                |            |         |                |

6. Substituting WQV (watershed inches) for runoff depth (Q), compute the water quality flow (WQF) from the following equation:

$$WQF = (q_u) * (A) * (Q)$$

$$WQV(cf) = (Af) * WQV(in) / 12$$

where: WQF = water quality flow (cfs)

WQV(in) = water quality volume (watershed inches)

$q_u$  = unit peak discharge (cfs/mi<sup>2</sup>/inch)

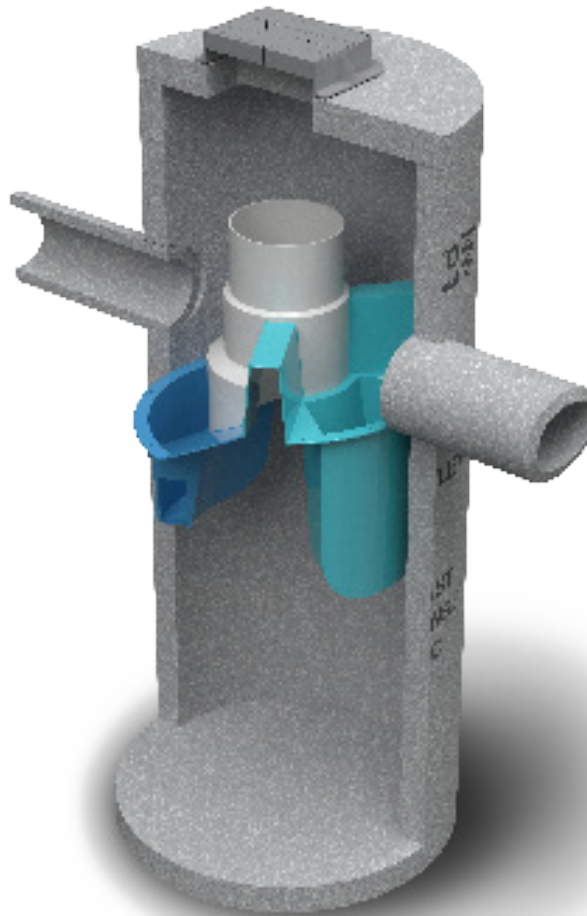
WQV = (Af)

A = drainage area (mi<sup>2</sup>)

Af = drainage area(sf) = Acres \* 43,560

Q = runoff depth (watershed inches)

| Structure Name | $q_u$ (csm/in) | A (miles <sup>2</sup> ) | Q (in) | WQF (cfs) | WQV (cf) |
|----------------|----------------|-------------------------|--------|-----------|----------|
| ST8            | 1000           | 0.00263                 | 0.870  | 2.28      | 5305.61  |
|                |                |                         |        |           |          |
|                |                |                         |        |           |          |
|                |                |                         |        |           |          |
|                |                |                         |        |           |          |



## Operation and Maintenance Manual

**First Defense® High Capacity and First Defense® Optimum**

---

Vortex Separator for Stormwater Treatment

## Table of Contents

|          |                                                                                                                                                                              |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>3</b> | <b>FIRST DEFENSE® BY HYDRO INTERNATIONAL</b> <ul style="list-style-type: none"><li>- INTRODUCTION</li><li>- OPERATION</li><li>- POLLUTANT CAPTURE AND RETENTION</li></ul>    |
| <b>4</b> | <b>MODEL SIZES &amp; CONFIGURATIONS</b> <ul style="list-style-type: none"><li>- FIRST DEFENSE® COMPONENTS</li></ul>                                                          |
| <b>5</b> | <b>MAINTENANCE</b> <ul style="list-style-type: none"><li>- OVERVIEW</li><li>- MAINTENANCE EQUIPMENT CONSIDERATIONS</li><li>- DETERMINING YOUR MAINTENANCE SCHEDULE</li></ul> |
| <b>6</b> | <b>MAINTENANCE PROCEDURES</b> <ul style="list-style-type: none"><li>- INSPECTION</li><li>- FLOATABLES AND SEDIMENT CLEAN OUT</li></ul>                                       |
| <b>8</b> | <b>FIRST DEFENSE® INSTALLATION LOG</b>                                                                                                                                       |
| <b>9</b> | <b>FIRST DEFENSE® INSPECTION AND MAINTENANCE LOG</b>                                                                                                                         |

---

**COPYRIGHT STATEMENT:** The contents of this manual, including the graphics contained herein, are intended for the use of the recipient to whom the document and all associated information are directed. Hydro International plc owns the copyright of this document, which is supplied in confidence. It must not be used for any purpose other than that for which it is supplied and must not be reproduced, in whole or in part stored in a retrieval system or transmitted in any form or by any means without prior permission in writing from Hydro International plc. First Defense® is a trademarked hydrodynamic vortex separation device of Hydro International plc. A patent covering the First Defense® has been granted.

**DISCLAIMER:** Information and data contained in this manual is exclusively for the purpose of assisting in the operation and maintenance of Hydro International plc's First Defense®. No warranty is given nor can liability be accepted for use of this information for any other purpose. Hydro International plc has a policy of continuous product development and reserves the right to amend specifications without notice.

# I. First Defense® by Hydro International

## Introduction

The First Defense® is an enhanced vortex separator that combines an effective and economical stormwater treatment chamber with an integral peak flow bypass. It efficiently removes total suspended solids (TSS), trash and hydrocarbons from stormwater runoff without washing out previously captured pollutants. The First Defense® is available in several model configurations to accommodate a wide range of pipe sizes, peak flows and depth constraints.

The two product models described in this guide are the First Defense® High Capacity and the First Defense® Optimum; they are inspected and maintained identically.

## Operation

The First Defense® operates on simple fluid hydraulics. It is self-activating, has no moving parts, no external power requirement and is fabricated with durable non-corrosive components. No manual procedures are required to operate the unit and maintenance is limited to monitoring accumulations of stored pollutants and periodic clean-outs. The First Defense® has been designed to allow for easy and safe access for inspection, monitoring and clean-out procedures. Neither entry into the unit nor removal of the internal components is necessary for maintenance, thus safety concerns related to confined-space-entry are avoided.

## Pollutant Capture and Retention

The internal components of the First Defense® have been designed to optimize pollutant capture. Sediment is captured and retained in the base of the unit, while oil and floatables are stored on the water surface in the inner volume (Fig. 1).

The pollutant storage volumes are isolated from the built-in bypass chamber to prevent washout during high-flow storm events. The sump of the First Defense® retains a standing water level between storm events. This ensures a quiescent flow regime at the onset of a storm, preventing resuspension and washout of pollutants captured during previous events.

Accessories such as oil absorbent pads are available for enhanced oil removal and storage. Due to the separation of the oil and floatable storage volume from the outlet, the potential for washout of stored pollutants between clean-outs is minimized.

## Applications

- Stormwater treatment at the point of entry into the drainage line
- Sites constrained by space, topography or drainage profiles with limited slope and depth of cover
- Retrofit installations where stormwater treatment is placed on or tied into an existing storm drain line
- Pretreatment for filters, infiltration and storage

## Advantages

- Inlet options include surface grate or multiple inlet pipes
- Integral high capacity bypass conveys large peak flows without the need for "offline" arrangements using separate junction manholes
- Long flow path through the device ensures a long residence time within the treatment chamber, enhancing pollutant settling
- Delivered to site pre-assembled and ready for installation

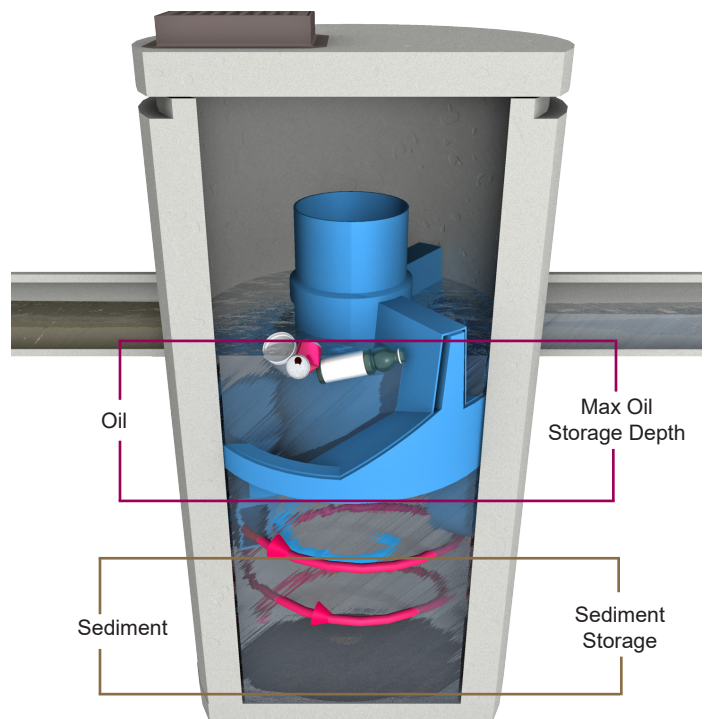


Fig. 1 Pollutant storage volumes in the First Defense®.

## II. Model Sizes & Configurations

The First Defense® inlet and internal bypass arrangements are available in several model sizes and configurations. The components have modified geometries allowing greater design flexibility to accommodate various site constraints.

All First Defense® models include the internal components that are designed to remove and retain total suspended solids (TSS), gross solids, floatable trash and hydrocarbons (Fig.2). First Defense® model sizes (diameter) are shown in Table 1.

## III. Maintenance

### First Defense® Components

1. Built-In Bypass

2. Inlet Pipe

3. Inlet Chute
4. Floatables Draw-off Port

5. Outlet Pipe

6. Floatables Storage
7. Sediment Storage

8. Inlet Grate or Cover

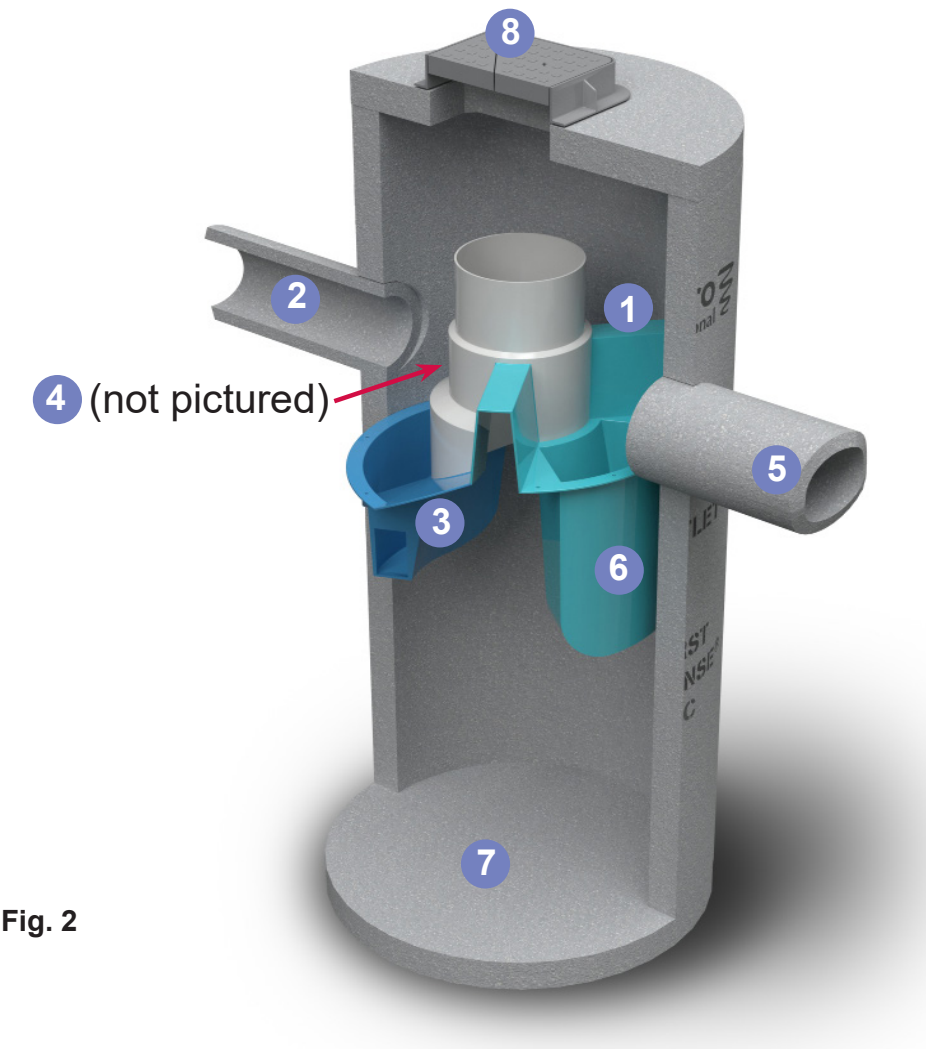


Fig. 2

Table 1

| First Defense®<br>Model Sizes |
|-------------------------------|
| (ft / m) diameter             |
| 3 / 0.9                       |
| 4 / 1.2                       |
| 5 / 1.5                       |
| 6 / 1.8                       |
| 7 / 2.1                       |
| 8 / 2.4                       |
| 10 / 3.0                      |

## Overview

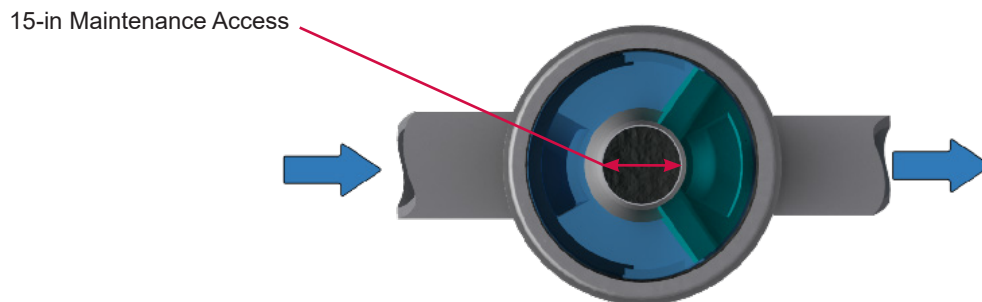
The First Defense® protects the environment by removing a wide range of pollutants from stormwater runoff. Periodic removal of these captured pollutants is essential to the continuous, long-term functioning of the First Defense®. The First Defense® will capture and retain sediment and oil until the sediment and oil storage volumes are full to capacity. When sediment and oil storage capacities are reached, the First Defense® will no longer be able to store removed sediment and oil.

The First Defense® allows for easy and safe inspection, monitoring and clean-out procedures. A commercially or municipally owned sump-vac is used to remove captured sediment and floatables. Access ports are located in the top of the manhole.

Maintenance events may include Inspection, Oil & Floatables Removal, and Sediment Removal. Maintenance events do not require entry into the First Defense®, nor do they require the internal components of the First Defense® to be removed. In the case of inspection and floatables removal, a vactor truck is not required. However, a vactor truck is required if the maintenance event is to include oil removal and/or sediment removal.

## Maintenance Equipment Considerations

The internal components of the First Defense® have a centrally located circular shaft through which the sediment storage sump can be accessed with a sump vac hose. The open diameter of this access shaft is 15 inches in diameter (Fig.3). Therefore, the nozzle fitting of any vactor hose used for maintenance should be less than 15 inches in diameter.



*Fig.3 The central opening to the sump of the First Defense® is 15 inches in diameter.*

## Determining Your Maintenance Schedule

The frequency of clean out is determined in the field after installation. During the first year of operation, the unit should be inspected every six months to determine the rate of sediment and floatables accumulation. A simple probe such as a Sludge-Judge® can be used to determine the level of accumulated solids stored in the sump. This information can be recorded in the maintenance log (see page 9) to establish a routine maintenance schedule.

The vactor procedure, including both sediment and oil / floatables removal, for First Defense® typically takes less than 30 minutes and removes a combined water/oil volume of about 765 gallons.

### Inspection Procedures

1. Set up any necessary safety equipment around the access port or grate of the First Defense® as stipulated by local ordinances. Safety equipment should notify passing pedestrian and road traffic that work is being done.
2. Remove the grate or lid to the manhole.
3. Without entering the vessel, look down into the chamber to inspect the inside. Make note of any irregularities. Fig.4 shows the standing water level that should be observed.
4. Without entering the vessel, use the pole with the skimmer net to remove floatables and loose debris from the components and water surface.
5. Using a sediment probe such as a Sludge Judge®, measure the depth of sediment that has collected in the sump of the vessel.
6. On the Maintenance Log (see page 9), record the date, unit location, estimated volume of floatables and gross debris removed, and the depth of sediment measured. Also note any apparent irregularities such as damaged components or blockages.
7. Securely replace the grate or lid.
8. Take down safety equipment.
9. Notify Hydro International of any irregularities noted during inspection.

### Floatables and Sediment Clean Out

Floatables clean out is typically done in conjunction with sediment removal. A commercially or municipally owned sump-vac is used to remove captured sediment and floatables (Fig.4).

Floatables and loose debris can also be netted with a skimmer and pole. The access port located at the top of the manhole provides unobstructed access for a vector hose to be lowered to the base of the sump.

### Scheduling

- Floatables and sump clean out are typically conducted once a year during any season.
- Floatables and sump clean out should occur as soon as possible following a spill in the contributing drainage area.



Fig.4 Floatables are removed with a vector hose

### Recommended Equipment

- Safety Equipment (traffic cones, etc)
- Crow bar or other tool to remove grate or lid
- Pole with skimmer or net (if only floatables are being removed)
- Sediment probe (such as a Sludge Judge®)
- Vector truck (flexible hose recommended)
- First Defense® Maintenance Log

### *Floatables and Sediment Clean Out Procedures*

1. Set up any necessary safety equipment around the access port or grate of the First Defense® as stipulated by local ordinances. Safety equipment should notify passing pedestrian and road traffic that work is being done.
2. Remove the grate or lid to the manhole.
3. Without entering the vessel, look down into the chamber to inspect the inside. Make note of any irregularities.
4. Remove oil and floatables stored on the surface of the water with the vactor hose or with the skimmer or net
5. Using a sediment probe such as a Sludge Judge®, measure the depth of sediment that has collected in the sump of the vessel and record it in the Maintenance Log (page 9).
6. Once all floatables have been removed, drop the vactor hose to the base of the sump. Vactor out the sediment and gross debris off the sump floor
7. Retract the vactor hose from the vessel.
8. On the Maintenance Log provided by Hydro International, record the date, unit location, estimated volume of floatables and gross debris removed, and the depth of sediment measured. Also note any apparent irregularities such as damaged components, blockages, or irregularly high or low water levels.
9. Securely replace the grate or lid.

## Maintenance at a Glance

| Inspection                 | <ul style="list-style-type: none"> <li>- Regularly during first year of installation</li> <li>- Every 6 months after the first year of installation</li> </ul> |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Oil and Floatables Removal | <ul style="list-style-type: none"> <li>- Once per year, with sediment removal</li> <li>- Following a spill in the drainage area</li> </ul>                     |
| Sediment Removal           | <ul style="list-style-type: none"> <li>- Once per year or as needed</li> <li>- Following a spill in the drainage area</li> </ul>                               |

NOTE: For most clean outs the entire volume of liquid does not need to be removed from the manhole. Only remove the first few inches of oils and floatables from the water surface to reduce the total volume of liquid removed during a clean out.



## First Defense® Installation Log

|                                       |               |
|---------------------------------------|---------------|
| HYDRO INTERNATIONAL REFERENCE NUMBER: |               |
| SITE NAME:                            |               |
| SITE LOCATION:                        |               |
| OWNER:                                | CONTRACTOR:   |
| CONTACT NAME:                         | CONTACT NAME: |
| COMPANY NAME:                         | COMPANY NAME: |
| ADDRESS:                              | ADDRESS:      |
| TELEPHONE:                            | TELEPHONE:    |
| FAX:                                  | FAX:          |

INSTALLATION DATE:     /     /

MODEL SIZE (CIRCLE ONE):     [3-FT]     [4-FT]     [5-FT]     [6-FT]     [7-FT]     [8-FT]     [10-FT]

INLET (CIRCLE ALL THAT APPLY):     GRATED INLET (CATCH BASIN)     INLET PIPE (FLOW THROUGH)

# First Defense® Inspection and Maintenance Log

[illegible]





## Stormwater Solutions

94 Hutchins Drive  
Portland, ME 04102

Tel: (207) 756-6200

Fax: (207) 756-6212

[stormwaterinquiry@hydro-int.com](mailto:stormwaterinquiry@hydro-int.com)

[www.hydro-int.com](http://www.hydro-int.com)

Turning Water Around...®

FD\_O+M\_K\_2105

# First Defense®

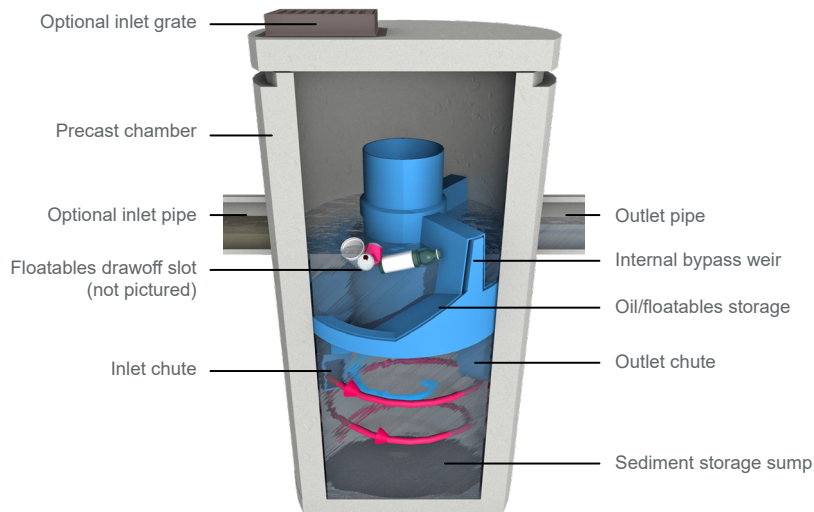
## Advanced Hydrodynamic Separator

### Product Summary

#### A Simple Solution for the Trickiest Sites

First Defense is a versatile stormwater separator with some of the highest approved flow rates in the United States. Engineers and contractors can save site space and reduce project costs by using the smallest possible footprint. It works with single or multiple inlet pipes and inlet grates. An internal bypass conveys infrequent peak flows directly to the outlet, efficiently capturing pollutants and preventing washouts.

### Features



Contaminated stormwater runoff enters the inlet chute from a surface grate and/or inlet pipe. The inlet chute introduces flow into the chamber tangentially to create a low energy vortex flow regime (magenta arrow) that directs sediment into the sump while oils, floating trash and debris rise to the surface.

Treated stormwater exits through a submerged outlet chute located opposite to the direction of the rotating flow (blue arrow). Enhanced vortex separation is provided by forcing the rotating

flow within the vessel to follow the longest path possible rather than directly from inlet to outlet.

Higher flows bypass the treatment chamber to prevent turbulence and washout of captured pollutants. An internal bypass conveys infrequent peak flows directly to the outlet eliminating the need for, and expense of, external bypass control structures. A floatables draw off slot functions to convey floatables into the treatment chamber prior to bypass.



### Benefits

#### Highest Flow Through the Smallest Footprint

##### » Smaller Footprint, Lower Costs

First Defense provides space-saving, easy-to-install surface water treatment in standard size chambers/manholes.

##### » Adapt to Site Limitations

Variable configurations will help you effectively slip First Defense into a tight spot. It also works well with large pipes, multiple inlet pipes and inlet grates.

##### » Reduce Installation Time & Costs

Every First Defense unit is delivered to site pre-assembled and ready for install.



### Applications

- » Stormwater treatment at the point of entry into the drainage line
- » Sites constrained by space, topography or drainage profiles with limited slope and depth of cover
- » Highways, parking lots, industrial areas and urban developments
- » Pre-treatment to ponds, storage systems, green infrastructure

### Maintenance

Easy vector hose access through the center shaft of the system makes for quick, simple sump cleanout while trash and floatables can be fished out from the surface with a net.

To ensure optimal performance, recommend Hydro International to your clients as the [preferred service and maintenance provider](#) .

## Sizing & Specifications

First Defense units are available in **six diameters** to fit standard chamber and manhole sizes. The dimensions below are common across all model numbers.

| Diameter | Peak Online Flow Rate | Maximum Pipe Diameter <sup>1</sup> | Typical Sediment Storage Capacity <sup>2</sup> | Minimum Distance from Outlet Invert to Top of Rim <sup>3</sup> | Standard Distance from Outlet Invert to Sump Floor |
|----------|-----------------------|------------------------------------|------------------------------------------------|----------------------------------------------------------------|----------------------------------------------------|
| (ft / m) | (cfs / L/s)           | (in / mm)                          | (yd <sup>3</sup> / m <sup>3</sup> )            | (ft / m)                                                       | (ft / m)                                           |
| 3 / 0.9  | 15 / 424              | 18 / 450                           | 0.4 / 0.3                                      | 2.0 - 2.5 / 0.61 - 0.76                                        | 3.71 / 1.13                                        |
| 4 / 1.2  | 18 / 510              | 24 / 600                           | 0.7 / 0.5                                      | 2.0 - 3.0 / 0.61 - 0.91                                        | 4.97 / 1.5                                         |
| 5 / 1.5  | 20 / 566              | 24 / 600                           | 1.1 / .84                                      | 2.0 - 3.7 / 0.61 - 1.13                                        | 5.83 / 1.5                                         |
| 6 / 1.8  | 32 / 906              | 30 / 750                           | 1.6 / 1.2                                      | 2.0 - 4.1 / 0.61 - 1.25                                        | 5.97 / 1.8                                         |
| 8 / 2.4  | 50 / 1415             | 48 / 1200                          | 2.8 / 2.1                                      | 2.4 - 5.4 / 0.73 - 1.65                                        | 7.40 / 2.2                                         |
| 10 / 3.0 | 50 / 1415             | 48 / 1200                          | 4.4 / 3.3                                      | 2.4 - 6.8 / 0.73 - 2.07                                        | 10.25 / 3.12                                       |

Hydro International offers First Defense units in **two versions** that conform to the performance requirements of different states' water quality regulations.<sup>4</sup>

| First Defense High Capacity Model Number | Typical TSS Treatment Flow Rates |               |
|------------------------------------------|----------------------------------|---------------|
|                                          | NJDEP Certified <sup>4</sup>     | 110µm         |
|                                          | (cfs / L/s)                      | (cfs / L/s)   |
| FDHC-3                                   | 0.84 / 23.7                      | 1.06 / 30.0   |
| FDHC-4                                   | 1.50 / 42.4                      | 1.88 / 53.2   |
| FDHC-5                                   | 2.35 / 66.2                      | 2.94 / 83.2   |
| FDHC-6                                   | 3.38 / 95.7                      | 4.23 / 119.8  |
| FDHC-8                                   | 6.00 / 169.9                     | 7.52 / 212.9  |
| FDHC-10                                  | 9.38 / 265.6                     | 11.75 / 332.7 |

| First Defense Optimum Model Number | NJDEP Certified Treatment Flow Rates <sup>4</sup> |
|------------------------------------|---------------------------------------------------|
|                                    | (cfs / L/s)                                       |
| FDO-3                              | 1.02 / 28.9                                       |
| FDO-4                              | 1.81 / 51.3                                       |
| FDO-5                              | 2.83 / 80.0                                       |
| FDO-6                              | 4.07 / 115.2                                      |
| FDO-8                              | 7.23 / 204.7                                      |
| FDO-10                             | 11.33 / 320.6                                     |

<sup>1</sup>Contact Hydro International when larger pipe sizes are required.

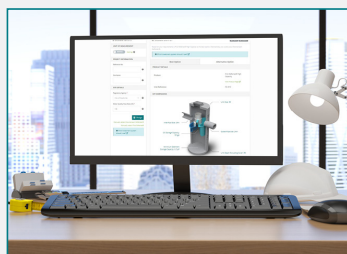
<sup>2</sup>Contact Hydro International when custom sediment storage capacity is required.

<sup>3</sup>These are guidelines only. Minimum distance is based on pipe diameter and headloss at assumed flow rates, contact Hydro for detailed design.

<sup>4</sup>NJDEP Certified / NJCAT Verified , based on one inlet pipe and no inlet grate.



**Also available in a screened configuration for Full Trash Capture!**



## Free Online Design Tool

This free online sizing tool will recommend the best separator, model size and online or offline configuration based on site-specific data entered by the user.

Upon completion, users have the option to submit the design to Hydro International for a free review by our engineering team.

Go to [hydro-int.com/sizing](https://hydro-int.com/sizing)  to access the tool.



📍 Hydro International, 94 Hutchins Drive, Portland, ME 04102

☎ Tel: (207) 756-6200

✉ Email: [stormwaterinquiry@hydro-int.com](mailto:stormwaterinquiry@hydro-int.com)

🌐 Web: [www.hydro-int.com/firstdefense](https://www.hydro-int.com/firstdefense) 

FD\_SS\_A\_2112

### Download Drawings:

→ [hydro-int.com/fddrawings](https://hydro-int.com/fddrawings) 

### Operation & Maintenance Manual:

→ [hydro-int.com/fd-om](https://hydro-int.com/fd-om) 

**MAINE DEPARTMENT OF ENVIROMENTAL PROTECTION**

**DIVISION OF WATERSHED MANAGEMENT**

**WITNESS REPORT**

**BY: JEFF DENNIS**

**DATE: NOVEMBER 12, 2004**

**Hydro International First Defense OK-110 Sand  
SSC (TSS) Removal Confirmation Test  
November 12, 2004**

Reported by Jeff Dennis  
Division of Watershed Management, DEP

On November 12, 2004 I witnessed a confirmation test of the ability of a 4 ft diameter First Defense® unit with an 8 inch inlet to remove OK-110 grade silica sand. The test was performed in the laboratory of the Hydro International office on Hutchins Drive in Portland, Maine. The target flow rate for the test was 320 gpm.

**Lab Set-Up**

The laboratory set-up for the test consists of a 23,300 gallon clean water storage reservoir from which water is pumped into an 8 in pipe which feeds water to a 4 ft diameter First Defense® unit. The pipe from the storage reservoir is fitted with a valved bypass to divert excess flows back to the storage reservoir, a butterfly valve along with a variable frequency drive for flow control, and an ISCO UniMag Magnetic Flowmeter. OK-110 sand is fed into the inflow pipe from an elevated 60 gal sand slurry barrel. The sand is kept in a relatively uniform suspension in the slurry tank using a propeller type mixer. Slurry is pumped through plastic tubing from the slurry tank into the inflow pipe by a peristaltic pump. An automatic sampler is located upstream of the slurry feed to collect background samples. Several feet downstream of the slurry feed in the inflow pipe there is a 6 inch T with a sluice gate for collection of inflow samples.

The outflow pipe from the First Defense® unit has a free-fall discharge back into the storage reservoir. Outflow samples are collected by passing the sample bottle through the free fall discharge into the reservoir.

**Test Procedure**

The target test flow for the test was 320 gpm. The mean water detention time in the system at this flow rate is 78 seconds. Outflow samples lagged inflow samples by this amount. The interval between samples for both the inflow and outflow samples was 60 seconds. Background samples were collected at the same time as inflow samples. Flow was observed throughout the test.

The flow rate was stabilized at around 300 gpm and the slurry feed pump started. The system was then allowed to reach equilibrium for a period in excess of four detention times, before the first inflow sample was taken. Outflow sampling commenced about 78 seconds later. Background sampling commenced prior to inflow sampling and continued throughout the test. Six sets of samples were taken.

Inflow, outflow and background samples were taken to the University of Maine Environmental Chemistry Lab for Suspended Sediment Concentration analysis. The analyses was performed by John Cangelosi.

## Results

Results of the test are presented in the attached tables. Inflow concentrations ranged from 189.1 mg/l to 299.8 mg/l. Outflow concentrations ranged from 12.6 mg/l to 17.3 mg/l. Background concentrations ranged between 0.9 and 1.9 mg/l.

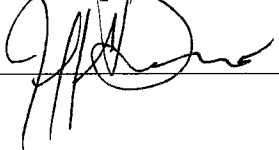
The removal efficiencies indicated by inflow/outflow pairs ranged from 93.3% up to 95.4%, with a mean of 94.2%. When adjusted for recycled background concentrations, efficiencies were slightly higher, from 94.0% to 95.7% with a mean of 94.7%.

Flow for the test varied from 262 gpm to 328 gpm with a mean of 290 gpm, slightly lower than the target flow rate of 320 gpm.

## Conclusions

All the paired sample removal efficiencies exceeded 80%, as did their mean whether or not they were adjusted for background concentrations, so it is very clear that at 290 gpm, a 4 ft diameter First Defense® unit can remove at least 80% of OK-110 grade silica sand, and seems to be able to remove more than 90% at this flow. Variation in paired removal efficiencies was low, and variation in inflow concentration was high, but still acceptable. Since removal efficiencies were so much higher than the required 80% and the flow for at least one pair exceeded 320 gpm, it is reasonable to conclude that, even though the mean flow was less than the target flow of 320 gpm, the unit can remove greater than 80% of OK-110 grade silica sand at the target flow rate of 320 gpm.

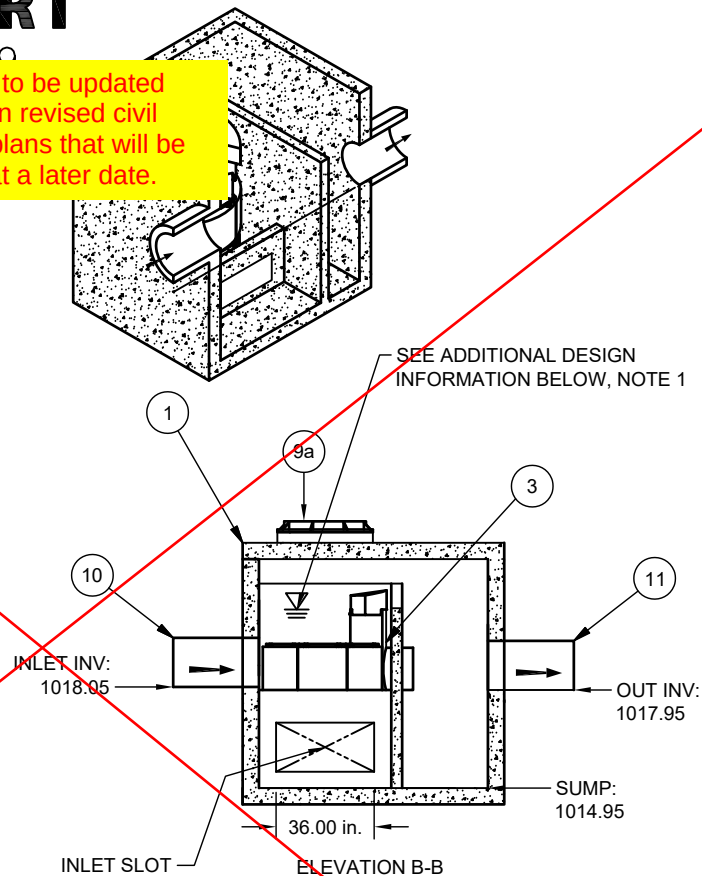
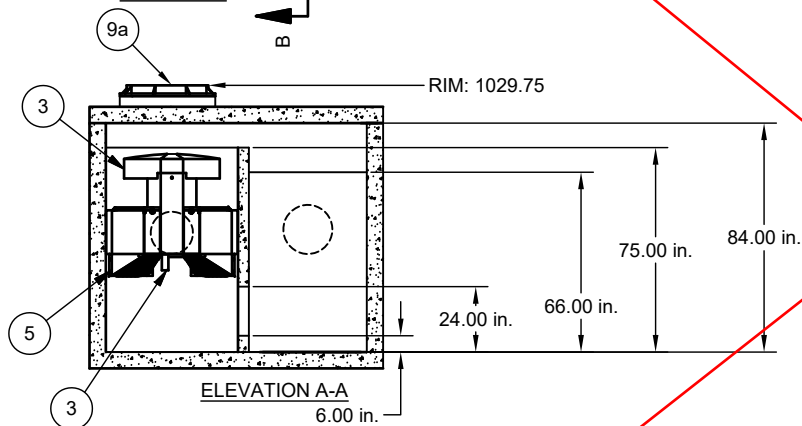
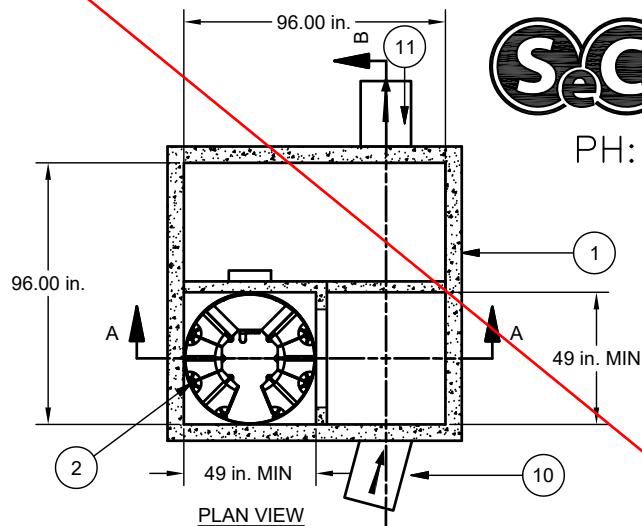
Therefore, the conclusion of this report is that the test performed on November 11, 2004, in substantial accordance with the Lab Testing Protocol, indicates that a 4 ft diameter First Defense® unit operating at an average flow rate of 320 gpm provides at least 80% removal of the specified OK-110 grade silica sand.

Signed:  Date: 7/27/05

# **SOUTHEAST CULVERT**

PH: 770-868-5500

Feature to be updated  
based on revised civil  
design plans that will be  
issued at a later date.



## **Notes**

1. MANHOLE WALL AND SLAB THICKNESSES ARE NOT TO SCALE.
2. CONTACT HYDRO INTERNATIONAL FOR A BOTTOM OF STRUCTURE ELEVATION PRIOR TO SETTING DOWNSTREAM DEFENDER MANHOLE.

## **REVISION HISTORY**

| REV             | BY  | DATE           | DESCRIPTION     |
|-----------------|-----|----------------|-----------------|
| B               | JLL | 2/02/15        | NOTES           |
| Date<br>1/15/26 |     |                | Scale<br>NTS    |
| Drawn<br>MHM    |     | Checked<br>MRJ | Approved<br>MRJ |

## **Title**

8-FT x 8-FT VAULT  
UP-FLO FILTER  
6 MODULES

ST 5

LIDL #1517  
NORCROSS, GA

**Hydro**  
International

Stormwater Solutions  
94 Hutchins Drive  
Portland, Maine 04102  
Tel: (207) 756-6200  
Fax: (207) 756-6212  
stormwaterinquiry@hydro-int.com

| Parts List |     |                                                  |               |
|------------|-----|--------------------------------------------------|---------------|
| ITEM       | QTY | DESCRIPTION                                      | SIZE          |
| 1          | 1   | PRECAST VAULT (BY HYDRO VIA PRECASTER)           | 96 in x 96 in |
| 2          | 6   | FILTER MODULE ()                                 |               |
|            | 6   | FILTER BAG SET                                   |               |
| 3          | 1   | OUTLET MODULE W/ BRACKET, BYPASS HOOD, DRAINDOWN |               |
| 4          | 0   | DUAL SUPPORT BRACKET W/ ANGLED SCREEN            |               |
| 5          | 6   | SINGLE SUPPORT BRACKET W/ ANGLED SCREEN          |               |
| 6          | 6   | BACKER PLATE FOR SINGLE SUPPORT                  |               |
| 7          | 1   | MOUNTING SPOOL                                   |               |
| 8          | 1   | ADDITIONAL DRAINDOWN                             |               |
| 9a         | 1   | FRAME AND COVERS                                 | 30in          |
| 10         | 1   | INLET PIPE (RCP)                                 | 24 in         |
| 11         | 1   | OUTLET PIPE (RCP)                                | 24 in         |

## **CAPACITIES:**

1. Minimum performance: 80% removal of Sil-Co-Sil 106 (d50 = 22 microns) at the peak treatment flow.
2. NJDEP peak treatment flow: 0.056 cfs/module \* OR 0.66 acres of imperviousness/module, whichever results in the greater number of modules.
3. Maximum number of modules: 7 \*\*

## **ADDITIONAL DESIGN INFORMATION:**

1. \* Normal operating W.S.E. is 2.46' above the outlet invert at the peak treatment flow of 0.056 cfs/module. For a given flow the head requirement can be reduced by adding additional filters.
2. \*\* Treatment flows that require more modules will require a larger vault design.
3. Media Type: CPZ

Any warranty made by Hydro International only applies to those items supplied by it. Hydro International does not accept and expressly disclaims any responsibility or liability for any structure, plant or equipment (or the performance thereof) designed, built, manufactured or supplied by any third-party. Hydro International has a policy of continuous product development and reserves the right to amend the specifications of any of its products or equipment at any time. Hydro International expressly disclaims any liability for the performance of its equipment (or any part thereof) used or made subject to conditions outside of the conditions set forth in Hydro International's design specifications. Hydro International owns the copyright in and to this drawing, which is supplied in confidence, and all intended recipients of the drawing, by their use thereof, agree to hold the drawing in confidence and not to use it for any purpose other than for which it was supplied and not reproduce, in whole or in part, the drawing or any of the equipment or structures depicted therein, without prior written permission of Hydro International.



## Up-Flo Filter Sizing - Annual Load Calculations

**Project Name:** Lidl #1517  
**Location:** Norcross, GA

Mean Annual Rainfall: 51 inches  
 % Runoff Capture 0.9

### Proposed Site Information:

| Land Type  |            | EMC<br>(mg/L) |
|------------|------------|---------------|
| Impervious | 1.29 acres | 100           |
| Pervious   | 0.42 acres | 100           |
| Total      | 1.71 acres |               |

I = Impervious cover (Percentage) 0.754  
 C = volumetric runoff coefficient =  $0.05 + 0.009(I)$  0.729

Runoff Volume (Vr) = (Mean Annual Rainfall)\*C\*(Total Acreage)\*(43,560 ft<sup>3</sup>/ 12 in / ft) \*(% Capture)  
 Runoff Volume (Vr): 207,688 ft<sup>3</sup>  
 Total 207,688 ft<sup>3</sup>

Annual Mass Load (M): 1,296 lbs  
 Total 1,296 lbs

Up-Flo Filter Design:

|                                        |                     |
|----------------------------------------|---------------------|
| Percent Removal                        | 80%                 |
| Annual Filtration Media Load           | 20%                 |
| Annual Sump Sedimentation              | 80%                 |
| Annual Filtered Load                   | 207 lbs             |
| Annual Mass Load per Filter Module     | 40 lbs              |
| Minimum Sump Storage Volume per module | 2.8 ft <sup>3</sup> |

|                                 |                          |
|---------------------------------|--------------------------|
| <b># Filter Modules</b>         | <b>5.18</b>              |
| <b>Min. Sump Storage Volume</b> | <b>15 ft<sup>3</sup></b> |

# Up-Flo® Filter

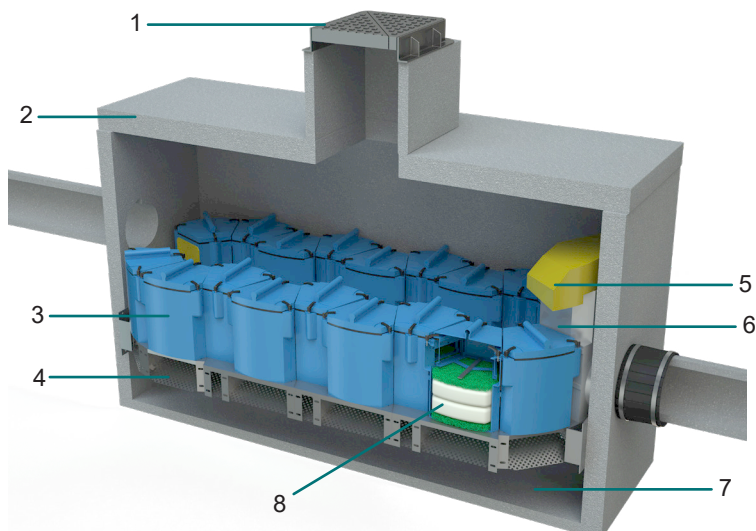
High Performance. Low Maintenance.

## Product Summary

### The Most Approved Filter in North America

Up-Flo® Filter utilizes a media blend so potent, it can remove up to 98% of TSS in a small surface area--media bags that weight less than 40lbs per module. This design enables high treatment of high flow with simple, low-labor maintenance.

Verified to remove 80% TSS but with demonstrated removal rates up to 98%, only the Up-Flo® Filter was developed in collaboration with the US Environmental Protection Agency and now has more approvals than any other stormwater filter in North America.



## Product Profile

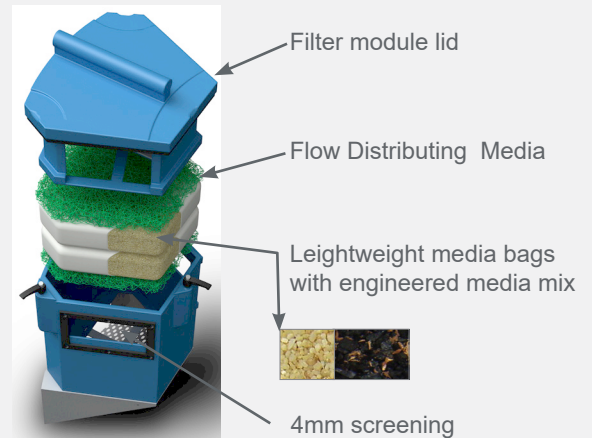
- |                                                     |                                         |
|-----------------------------------------------------|-----------------------------------------|
| 1. Inlet grate (pictured) or Inlet Pipe (not shown) | 5. Bypass Hood/Siphon                   |
| 2. Precast Filtration Chamber                       | 6. Outlet Module with Drain Down Filter |
| 3. Filter Module                                    | 7. Pollutant Storage Sump               |
| 4. 4mm Screening                                    | 8. Media bags                           |

## Applications

- » Sites requiring 80% TSS removal
- » Removal of sediment, nutrients and metals from runoff
- » Source control for redevelopment or new construction
- » Treatment downstream of detention systems
- » Sites operating under industrial or multi-sector general permits
- » Protection for groundwater recharge systems

## How it Works

### Proven to Remove up to 98% of TSS!



### Three Treatment Levels for Optimal Removal Performance.

**Stage 1: Pretreatment.** Oil and floatables rise to the surface while sediment settles in the sump. **Screening:** Flow is directed upward through an angled screen to remove debris before entering the filter module.

**Stage 2: Screening.** Flow is directed upward through an angled screen to remove debris before entering the filter module.

**Stage 3: High-rate Filtration.** Water flows upwards through engineered media bags (see **Fig.2**) before leaving the outlet module to be discharged through the outlet pipe.

During peak flows, excess water is siphoned through the yellow bypass hood which also prevents the escape of oil and trash. As water levels return to normal, captured pollutants are washed off media bags, preventing blinding and prolonging media life.

## Benefits

### High Performance. Low Maintenance.

- » More regulatory approvals than any other filter in North America
- » Proven to remove up to 98% of TSS
- » Reduce vault height by designing with zero drop between inlet and outlet (exceptions apply).
- » Light-weight media bags make for quick, easy, maintenance.



**Stormwater Solutions**

→ [hydro-int.com/up-flo-filter](http://hydro-int.com/up-flo-filter)

## Product Profile

### Zero Drop Means Shorter Vaults!

Save space and cost on vault height by using a filter that can be designed with zero drop from inlet to outlet. Contact Hydro International at [stormwaterinquiry@hydro-int.com](mailto:stormwaterinquiry@hydro-int.com) to confirm feasibility for your site.

| Configuration     | Treatment Chamber  | Maximum Filter Modules | Maximum Treatment Flow Rate | Minimum Height (Sump to Rim) | Sump Depth | Maximum Pipe Diameter | Standard Inlet to Outlet Drop | Minimum Inlet to Outlet Drop | Operating Head |
|-------------------|--------------------|------------------------|-----------------------------|------------------------------|------------|-----------------------|-------------------------------|------------------------------|----------------|
|                   | (ft/m)             | (No.)                  | (cfs/L/s)                   | (ft/m)                       | (ft/m)     | (in/mm)               | (ft/m)                        |                              | (ft/m)         |
| Round Manhole     | 4 / 1.2            | 6                      | 0.336                       | 7.5 / 2.29                   | 3.0 / 0.91 | 15 / 375              | 0.8 / 0.24                    | 0.0*                         | 2.5/ 0.76      |
| Rectangular Vault | 4 x 6 / 1.2 x 1.8  | 8                      | 0.448                       | 6.5 / 1.98                   | 2.0 / 0.60 | 24 / 609              |                               |                              |                |
|                   | 4 x 8 / 1.2 x 2.4  | 14                     | 0.784                       |                              |            |                       |                               |                              |                |
|                   | 4 x 10 / 1.2 x 3.0 | 18                     | 1.008                       |                              |            |                       |                               |                              |                |
|                   | 8 x 12 / 2.4 x 3.7 | 40                     | 2.24                        |                              |            |                       |                               |                              |                |
|                   | 8 x 21 / 2.4 x 6.4 | 76                     | 4.256                       |                              |            |                       |                               |                              |                |

\*Some filter projects can be designed with zero drop from inlet to outlet. Please consult with a Hydro International engineer before

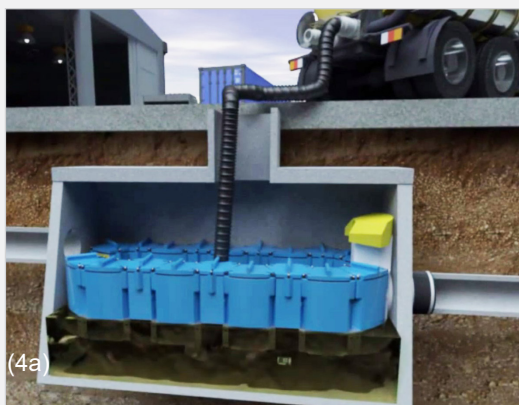
## Maintenance

### The ONLY filter with lightweight media bags for easy removal and maintenance!

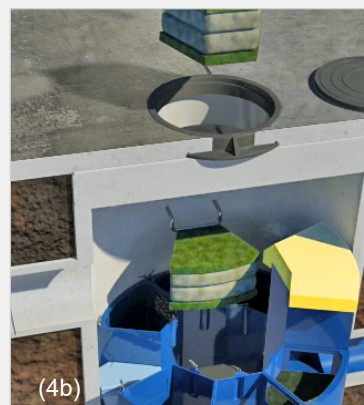
**Easy:** Maintenance is simple with easy access to the sump and replaceable media packs. A vacuum truck is used to remove sediment and debris from the sump (Fig.4a).

**Lightweight:** Unlike other filter systems whose media cartridges weigh upwards of 250lbs, our lightweight media bags can be manually replaced without removing the entire module (Fig.4b).

**Need Help?** Our prompt, friendly service teams are happy to take the dirty work off your hands. Call us for a quote at **1 (800) 848-2706**.



Sediment is removed with a standard vacuum truck.



Media bags are removed manually without use of heavy equipment.



📍 Hydro International, 94 Hutchins Drive, Portland, ME 04102

☎ Tel: (207) 756-6200

✉ Email: [stormwaterinquiry@hydro-int.com](mailto:stormwaterinquiry@hydro-int.com)

🌐 Web: [www.hydro-int.com/up-flo-filter](http://www.hydro-int.com/up-flo-filter)

UFF\_CPZ\_A\_SS\_2105

### Download Drawings!

→ [hydro-int.com/uff-drawings](http://hydro-int.com/uff-drawings)

### Access the Operation & Maintenance Manual

→ [hydro-int.com/uff-om](http://hydro-int.com/uff-om)



## Operation and Maintenance Manual

### Stormwater Solutions

#### Up-Flo® Filter

#### Filtration System for Stormwater Treatment

94 Hutchins Drive  
Portland, ME 04102

Tel: (207) 756-6200  
Fax: (207) 756-6212  
[stormwaterinquiry@hydro-int.com](mailto:stormwaterinquiry@hydro-int.com)

[www.hydro-int.com](http://www.hydro-int.com)

## TABLE OF CONTENTS

|    |                                                                                                                                                                                                                                                                                                                                                                         |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3  | Up-Flo® Filter <ul style="list-style-type: none"><li>- Overview</li><li>- General Product Description</li><li>- Typical Configurations</li><li>- Maintenance Services</li></ul>                                                                                                                                                                                         |
| 4  | Operation <ul style="list-style-type: none"><li>- Introduction</li><li>- Pollutant Capture</li><li>- Reduced Clogging</li><li>- Overflow Protection</li><li>- Best Practices</li><li>- Damage Due to Lack of Maintenance</li></ul>                                                                                                                                      |
| 5  | Inspection & Maintenance <ul style="list-style-type: none"><li>- Overview</li><li>- First-Year Monitoring</li><li>- Inspection</li><li>- Maintenance Activities Not Requiring Man Entry - Floatables, Oil and Sump Cleanout</li><li>- Maintenance Activities Requiring Man Entry - Replacement of Media Packs and Drain Down Filter</li><li>- Solids Disposal</li></ul> |
| 13 | Up-Flo® Filter Installation Log                                                                                                                                                                                                                                                                                                                                         |
| 14 | Up-Flo® Filter Inspection Log                                                                                                                                                                                                                                                                                                                                           |
| 16 | Up-Flo® Filter Maintenance Log                                                                                                                                                                                                                                                                                                                                          |

## IMPORTANT - ORDER REPLACEMENT PARTS FOR MAINTENANCE - IMPORTANT

Annual maintenance requires replacement of the Media Packs and the Drain Down Filter. Contact Hydro International to order replacements. Allow 2-4 weeks for delivery.

Office hours Monday thru Friday 8:00 A.M. to 5:00 P.M. EST

Toll free: 1-888-382-7808

Phone: 207-756-6200

Fax: 207-756-6212

Email: [services@hydro-int.com](mailto:services@hydro-int.com)

**COPYRIGHT STATEMENT:** The contents of this manual, including the drawings and specifications contained herein or annexed hereto, are intended for the use of the recipient to whom the document and all associated information are directed. Hydro International plc owns the copyright of this document (including any drawings or graphics), which is supplied in confidence. It must not be used for any purpose other than that for which it is supplied and must not be reproduced, in whole or in part, stored in a retrieval system or transmitted in any form or by any means without prior permission in writing from Hydro International plc. Up-Flo® Filter is a trademarked filtration device of Hydro International plc. A patent covering the Up-Flo® Filter has been granted.

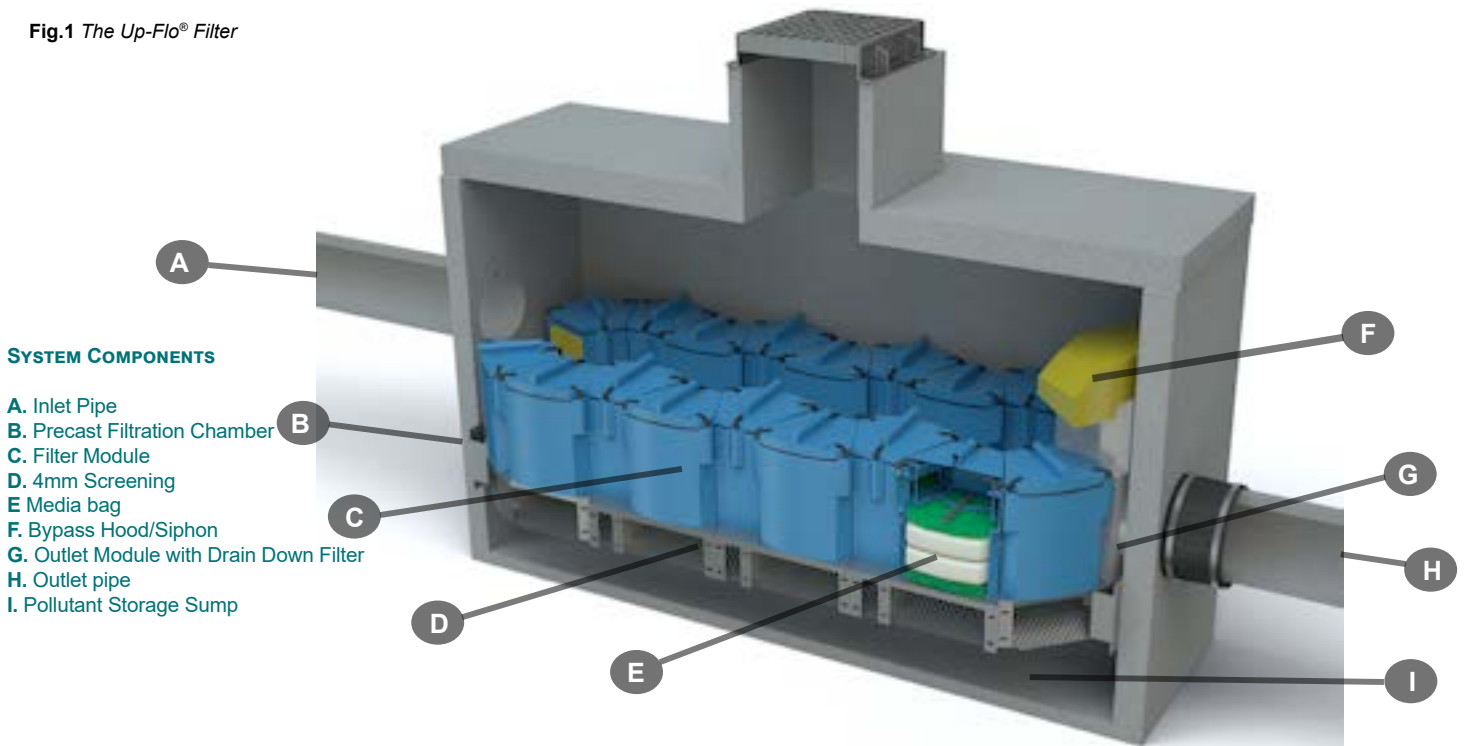
**DISCLAIMER:** Information and data contained in this manual is exclusively for the purpose of assisting in the operation and maintenance of Hydro International plc's Up-Flo® Filter. No warranty is given nor can liability be accepted for use of this information for any other purpose. Hydro International plc have a policy of continuous product development and reserve the right to amend specifications without notice.

## OVERVIEW & PRODUCT DESCRIPTION

The Up-Flo® Filter is a modular high-rate stormwater filtration device designed to capture trash, oil, sediment and remove fine pollutants such as dissolved and particulate metals and nutrients from stormwater runoff. Designed with efficiency, longevity and upkeep in mind, this high performance, low maintenance filter option that offers higher loading rates and longer media life for higher quality stormwater for longer periods between servicings.

In general, a minimum of two inspections are required per year to monitor sediment and gross pollutant accumulations. In order to achieve an annual TSS removal rate of 80% for the Up-Flo® Filter, the minimum maintenance frequency specified in the maintenance section for replacement of the Media Pack and removal of accumulated sediment from the sump is mandatory.

**Fig.1** The Up-Flo® Filter



## PRODUCT CONFIGURATIONS



**Fig.2** The Up-Flo® Filter is installed in a) 4-ft (1.2m) round manholes or b) in rectangular precast vaults. Both configurations have a wide central opening in the Up-Flo® Filter.

## HYDRO MAINTENANCE SERVICES

Hydro International has been engineering stormwater treatment systems for over 30 years. We understand the mechanics of removing pollutants from stormwater and how to keep systems running at an optimal level.

### NOBODY KNOWS OUR SYSTEMS BETTER THAN WE DO



### AVOID SERVICE NEGLIGENCE

Sanitation services providers not intimately familiar with stormwater treatment systems are at risk of the following:

- Inadvertently breaking parts or failing to clean/replace system components appropriately.
- Charging you for more frequent maintenance because they lacked the tools to service your system properly in the first place.
- Billing you for replacement parts that might have been covered under your Hydro warranty plan
- Charging for maintenance that may not yet have been required.

### LEAVE THE DIRTY WORK TO US

Trash, sediment and polluted water is stored inside treatment systems until they are removed by our team with a vactor truck. Sometimes teams must physically enter the system chambers in order to prepare the system for maintenance and install any replacement parts. Services include are are not limited to:

- Solids removal
- Removal of liquid pollutants
- Replacement media installation (when applicable)



## BETTER TOOLS, BETTER RESULTS

Not all vacor trucks are created equal. Appropriate tools and suction power are needed to service stormwater systems appropriately. Companies who don't specialize in stormwater treatment won't have the tools to properly clean systems or install new parts.



## SERVICE WARRANTY

Make sure you're not paying for service that is covered under your warranty plan. Only Hydro International's service teams can identify tune-ups that should be on us, not you.

## TREATMENT SYSTEMS SERVICED BY HYDRO:

- Stormwater filters
- Stormwater separators
- Baffle boxes
- Biofilters/biorention systems
- Storage structures
- Catch basins
- Stormwater ponds
- Permeable pavement



SAVE TIME & MONEY: CALL HYDRO FOR A QUOTE

**1 (800) 382-7808**

LEARN MORE AT [HYDRO-INT.COM/SERVICE](http://HYDRO-INT.COM/SERVICE)

## OPERATION

### INTRODUCTION

The Up-Flo® Filter operates on simple fluid hydraulics. It is self-activating, has no moving parts, no external power requirements and is fabricated with durable non-corrosive components. Personnel are not required to operate the unit and maintenance is limited to periodic inspections, sediment and floatables removal, Media Pack replacement and Drain Down Filter replacement.

### POLLUTANT CAPTURE

The Up-Flo® Filter is designed to operate as a “treatment train” by incorporating multiple treatment technologies into a single device. Trash and gross debris are removed by sedimentation and screening before they are introduced to the filtration media, preventing surface blinding of the filter media. The Up-Flo® Filter is a wet-sump device. Between storm events, oil and floatables are stored on the water surface separate from the sediment storage volume in the sump (see **Fig.1**). The high-capacity bypass siphon acts as a floatables baffle to prevent washout of captured floatable pollutants during high intensity events.

### REDUCED CLOGGING

The Up-Flo® Filter has been designed to minimize the occurrence of clogging and blinding and employs a unique Drain Down Filter that allows the water level in the chamber to drop below the filter media between events. The Drain Down Filter mechanism creates a reverse flow that flushes captured pollutants off the surface of the Media Bag, helping to prevent blinding. By allowing the water to drain out, the Drain Down Filter also reduces the weight of the Media Bags. This makes the bags easier and safer to remove during maintenance operations.

### OVERFLOW PROTECTION

The Angled Screens are designed to prevent ragging and blinding and are situated below the Filter Modules, sheltering them from the direct path of the influent. Coarse debris settles in the sump before the runoff flows up through the screens, protecting them from blinding. In the unlikely event of a blockage, the high capacity siphonic Bypass Hood is designed to convey high enough flow to minimize the risk of large storm creating upstream flooding.

### BEST PRACTICES

Good housekeeping upstream of the Up-Flo® Filter can significantly extend Media Bag life. For example, sweeping paved surfaces, collecting leaves and grass trimmings, and protecting bare ground from erosion will reduce loading to the system. Media Packs should not be installed in the Filter Modules until construction activities are complete and site stabilization is effective.

### DAMAGE DUE TO LACK OF MAINTENANCE

Delayed maintenance would result in clogged Media Bags and/or blinded Angled Screens. In that situation, the Up-Flo® Filter would go into bypass and there would be no treatment of the incoming stormwater. Because the Bypass Weir can easily convey all of the flow to the Outlet Module, there would be no lasting damage to the system. Replacement of the Media Bags and removal of sediment from the sump would restore the Up-Flo® Filter to its original treatment efficiency. Establishing and adhering to a regular maintenance schedule ensures optimal performance of the system.



Fig.3 a) The water level in a properly functioning Up-Flo® Filter will drain down to the base of the Filter Modules. b) When the Drain Down Filter becomes clogged, the base of the Filter Modules will be submerged in standing water. Note, above right, that the Drain Down Filter is submerged in standing water.

## INSPECTION & MAINTENANCE

### OVERVIEW

The Up-Flo® Filter protects the environment by removing a wide range of pollutants from stormwater runoff. Periodic removal of these captured pollutants is essential to the proper functioning of the Up-Flo® Filter.

Maintenance activities can be categorized as those that may be performed from outside the Up-Flo® vessel and those that are performed inside the vessel. Maintenance performed from outside the modules includes removal of floatables and oils that have accumulated on the water surface and removal of sediment from the sump. Maintenance performed inside the vessel includes removal and replacement of Media Bags, Flow Distribution Media and the Drain Down Filter. A vactor truck is required for removal of oils, water, sediment, and to completely pump out the vessel to allow for maintenance inside. If you are not using Hydro International or a trained service provider you must follow OSHA Confined Space Entry procedures when entering the Up-Flo® vessel.

The Up-Flo® Filter design has a wide central opening between the Filter Modules for easy access to all of the components (see **Fig.3**). In the case of inspection and floatables removal, a vactor truck is not required. Otherwise, a vactor truck is normally required for oil removal, removal of sediment from the sump, and replacement of the Media Packs and Drain Down Filter. In most cases, entry into the Up-Flo® Filter vessel is required for replacement of the Media Packs and Drain Down Filter.

**The minimum required frequency for replacement of the Media Pack is annually**, whereas the minimum required frequency for removal of accumulated sediment from the sump is dependent on the Up-Flo® Filter configuration. Configurations with a larger sediment storage volume per module will require less frequent removal of accumulated sediment. Regardless, whenever sediment depth in the sump is found to be greater than 16 inches, sediment removal is required.



**AT A MINIMUM, MEDIA BAGS MUST BE REPLACED AT LEAST ONCE A YEAR.**

Fig.4 a) A new Media Bag of Hydro Filter Sand. b) A spent media bag of Hydro Filter Sand.

## MAKE SURE YOUR SYSTEM WAS INSTALLED CORRECTLY

### First Year Inspection and Maintenance

The frequency of inspection and maintenance can be determined in the field after installation. The frequency of ongoing maintenance needs is based on site characteristics such as contributing area, types of surfaces (e.g., paved and/or landscaped), site activities (e.g., short-term or long-term parking), and other site maintenance (e.g., sanding and sweeping). At a minimum, inspection and maintenance should be conducted at intervals of no more than six months during the first year of operation. Maintenance personnel should observe and record pollutant accumulations during the first year of service in order to benchmark the maintenance intervals that will later be established for the site. Pollutant accumulations should be measured or monitored using the following procedures:

- **Measurement of sediment depth in the sump:** A minimum of 8 inches (20 cm) should separate the Drain Down Filter inlet from stored sediment in the sump in order to minimize sediment migration into the Drain Down Filter. A simple probe, such as the Sludge-Judge®, can be used to determine the depth of the solids in the sump. In a typical 4-ft (1.2m) diameter manhole installation, the sediment depth should be no more than 16 inches (41 cm).
- **Maintenance personnel should then enter the structure, remove the Media Pack from one of the Filter Modules, and weigh the Media Bags.** Media Bags with a wet weight of approximately 40 lbs (18 kg) or more are an indication that the filter media has become full and that the Media Packs in all of the Filter Modules will require replacement (Fig.4). Minimum filtration rate is generally reached when the Media Bags have accumulated approximately 20 lbs (9 kg) of sediment. Determining the amount of accumulated sediment will be accomplished by removing both of the Media Bags from one of the Media Packs and weighing the bags separately. Since a new Media Bag weighs approximately 30 lbs (14 kg) wet, the difference in weight will approximately equal the weight of solids that have accumulated in the bag. A spent Media Bag weighs approximately 50 lbs (23 kg) wet.
- **Measurement of oil layer on water surface:** Since water in the Up-Flo® vessel drains down to an elevation below the bottom of the Filter Modules when the system is idle, the amount of accumulated oil must be minimized so that oil is not entrained in the Media Pack when stormwater begins to fill the vessel at the start of a storm event. Oil accumulation should be limited to 1.5 inches (4 cm) or less. Probes can be used to measure oil thickness.
- **Monitoring for Drain Down Filter clogging:** The water level in the Up-Flo® Filter should be monitored to ensure that the Drain Down Filter is operating properly. The Drain Down Filter is designed to lower the water level in the Up-Flo® vessel to an elevation below the bottom of the Filter Modules between storm events. Periodically conduct an inspection one to two days after a storm event during the first year of operation. Approximately 36 hours after a 1-in (2.5-cm) rainfall, the water level inside the vessel should have dropped to a point where it is equal with the base of the Filter Modules. If the water level has not reached that point, then the Drain Down Filter has either become clogged or blinded by trash or debris (Fig.5 a and b). If there is no evidence of trash or debris around the Drain Down Filter inlet, then it has likely become clogged with particles.
- **Monitoring for slime and debris covering the Flow Distribution Media or Angled Screens:** After removal of the Media Bags, the bottom Flow Distribution Media should be removed and inspected to determine if it is coated with slime or debris. Similarly, the Angled Screen should be inspected for blockages and ragging.

## FIND OUT HOW FREQUENTLY YOUR SYSTEM NEEDS MAINTENANCE

Monitoring for floatables on the water surface: Similar to oil, the amount of accumulated floatables must be minimized to prevent trash and loose debris from becoming trapped on the Angled Screens when stormwater begins to fill the Up-Flo® vessel at the start of a storm event. Visual inspection is adequate to determine the amount of floatables. Floatables should be removed before they form a mat on the surface of the water.

The solids loading rate in the sump will be calculated by measuring the sediment depth in the sump and dividing the depth by the correlating interval of time since the sump was last cleaned. Similarly, starting with fresh Media Bags, the solids loading rate in the Media Packs will be calculated by weighing the Media Bags and dividing the weights by the correlating interval of time since they were installed. The wet weight of the heaviest bag will be used to determine the loading rate. As previously mentioned, a spent Media Bag weighs approximately 50 lbs (23 kg) wet. The spent Media Bag weight estimate was based on calculations of sediment loading in an Up-Flo® Filter that was run to exhaustion during laboratory testing.

The rate of oil accumulation will be calculated by measuring the thickness of the oil layer and dividing the thickness by the correlating interval of time since the sump was last cleaned. Ordinarily, oil thickness will not be measurable unless a spill has occurred. Consequently, any oil will typically be removed along with water when cleaning the sump.

Monitoring the Drain Down Filter for clogging, monitoring the Flow Distribution Media and Angled Screens for slime and debris, and monitoring the accumulation of floatables will provide an estimate of how long the Up-Flo® Filter can operate before its performance can become impaired by one of these factors.

### Routine Inspection and Maintenance

After completion of the first year of operation, determining and then following the established inspection and maintenance intervals will keep pollutant loadings within their respective limits. Removal of oils and floatables, replacement of the Drain Down Filter, replacement of Flow Distribution Media (see Fig.9, pg 11), and cleaning of Angled Screens will occur at the same frequency as cleaning of the sump and replacement of Media Bags unless the first year of operation indicates otherwise. Keeping to the established maintenance intervals will keep treatment flow rates at, or above, the design flow rate. Typically, annual maintenance is adequate.

In addition to scheduled maintenance, occasional checks for Up-Flo® Filter clogging can be performed by removing the manhole cover during a storm, monitoring the water level in the manhole or vault, and determining whether the filter is in bypass. A properly-sized filter (on-line or off-line) that is in bypass during a storm that is producing runoff at, or below, the filter's design filtration rate needs maintenance.

**DON'T WANT TO GO IT ALONE? CALL HYDRO AND WE'LL TAKE CARE OF INSPECTION, REPLACEMENT MEDIA AND CLEANOUT.**

**CALL 1 (888) 382-7808 FOR A QUOTE**

# INSPECTION & MAINTENANCE

## ROUTINE INSPECTION

Inspection is a simple process that requires monitoring pollutant accumulations. Maintenance crews should be familiar with the Up-Flo® Filter and its components prior to inspection.

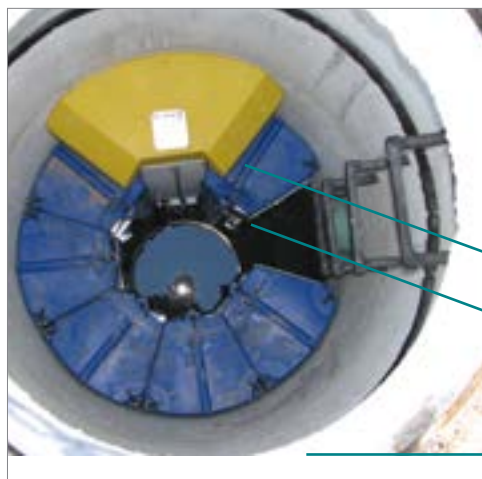
**THE FOLLOWING INSTRUCTIONS ARE INTENDED FOR NON-HYDRO MAINTENANCE SERVICE PROVIDERS AND/OR THOSE INTENDING TO MAINTAIN THEIR OWN UP-FLO® FILTER:**

### SCHEDULING

- Inspection may be conducted during any season of the year but should occur shortly after a predicted rainfall to ensure components are operating properly.

### NECESSARY EQUIPMENT

- Safety Equipment and Personal Protective Equipment (traffic cones, work gloves, etc.)
- Scale to measure the weight of the Media Bags
- Crow bar to remove grate or lid
- Pole with skimmer or net
- Sediment probe (such as a Sludge-Judge®)
- Hydro International Up-Flo® Filter Maintenance Log
- Trash bags for removed floatables



Bypass siphon sits evenly on Outlet Module.

Standing water level is no higher than the base of the Filter Module. The Drain Down Filter will be visible if the water level is correct.

Filter Module Lids are closed.

## ROUTINE INSPECTION PROCEDURES

1. Set up any necessary safety equipment (such as traffic cones) to provide access to the Up-Flo® Filter. Safety equipment should notify passing pedestrian and road traffic that work is being done.
2. Remove the grate or lid to the manhole or vault.
3. Without entering the vessel, look down into the chamber to inspect the inside and to determine whether the high-water level indicator has been activated. Make note of any irregularities. See Fig.6 for a typical Inspection View.
4. Without entering the vessel, use the pole with the skimmer net to remove floatables and loose debris from the chamber.
5. Using a sediment probe such as a Sludge-Judge®, measure the depth of sediment that has collected in the sump of the vessel. Maximum sediment depth is 16 inches (41 cm).
6. If the high-water level indicator has been activated after two consecutive storms, remove the Filter Module lid by turning the cam latch and remove the Filter Media Pack (*refer to page 11 Replacement Procedures*). Weigh the Media Bags from one or two modules. Media Bags should be replaced if the wet weight exceeds 40 lbs (18 kg).
7. On the Maintenance Log provided by Hydro International, record the date, unit location, estimated volume of floatables and gross debris removed, and the depth of sediment measured. Also note any apparent irregularities such as damaged components or a high standing water level (see Fig.6 for the standard standing water level).
8. Securely replace the grate or lid.
9. Remove safety equipment.
10. Contact Hydro International at (800) 848-2706 to discuss any irregularities noted during inspection.

Fig.6 Inspection view of the Up-Flo® Filter.

## ROUTINE MAINTENANCE

Maintenance activities are grouped into two categories:

- **Activities *Not Requiring Man Entry Into the Up-Flo® Filter***  
These activities include floatables removal, oil removal and removal of sediment from the sump.
- **Activities *Requiring Man Entry Into the Up-Flo® Filter***  
Media Pack replacement and Drain Down Filter replacement.

Maintenance intervals are determined from monitoring the Up-Flo® Filter during its first year of operation. Depending on the site, some maintenance activities may have to be performed on a more frequent basis than others. In the case of floatables removal, a vactor truck is not required. Floatables and loose debris can be netted with a skimmer and pole.

A vactor truck is normally required for oil removal, removal of sediment from the sump, and to dewater the vessel for replacement of the Media Packs and Drain Down Filter (Fig.7). All inspection and maintenance activities would be recorded in an Inspection and Maintenance Log.

Completion of all the maintenance activities for a typical 4-ft (1.2m) diameter manhole installation takes less than one hour. Approximately 360 gallons of water and up to 0.6 yd<sup>3</sup> (0.5 m<sup>3</sup>) of sediment may be removed in the process. In an installation equipped with six Filter Modules, 12 Media Bags (2 bags per module) would be removed and replaced. Assuming a spent Media Bag weight of 50 lbs (23 kg), up to 600 lbs (272 kg) of spent Media Bags would be removed. All consumables, including Media Bags, Flow Distribution Media, and replacement Drain Down Filters are supplied by Hydro International.

The access port located at the top of the manhole provides unobstructed access for a vactor hose and/or skimmer pole to be lowered to the base of the sump.

## MAINTENANCE ACTIVITIES NOT REQUIRING MAN ENTRY

These activities include floatables removal, oil removal and removal of sediment from the sump.

## SCHEDULING

- Floatables and sump cleanout may typically be done during any season of the year - before and after rainy season
- Floatables and sump cleanout should occur as soon as possible following a contaminated spill in the contributing drainage area

## RECOMMENDED EQUIPMENT

- Safety Equipment (traffic cones, etc)
- Crow bar to remove grate or lid
- Pole with skimmer or net (if only floatables are being removed)
- Sediment probe (such as a Sludge-Judge®)
- Vactor truck (flexible hose preferred)
- Pressure nozzle attachment or other screen-cleaning device

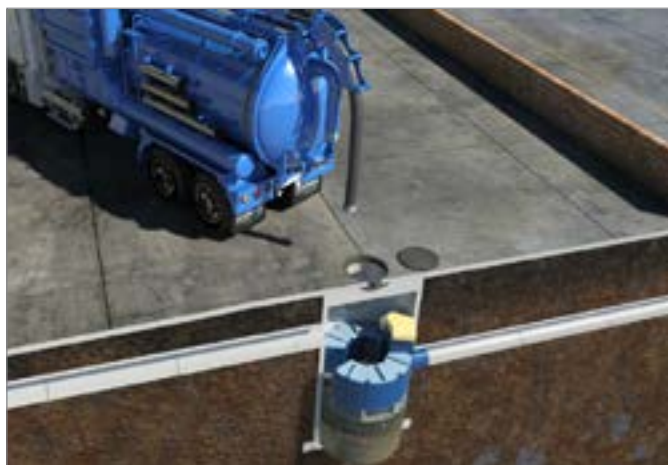


Fig.7 Sediment is removed from the sump with a vactor hose. Man entry is not required for this step.

**NO MAN ENTRY REQUIRED: FLOATABLES, OIL AND SEDIMENT:**

1. Set up any necessary safety equipment (such as traffic cones) around the access of the Up-Flo® Filter. Safety equipment should notify passing pedestrian and road traffic that work is being done.
2. Remove the grate or lid to the manhole or vault.
3. Without entering the vessel, look down into the chamber to inspect the inside. Make note of any irregularities.
4. If the standing water level in the sump is above the base of the Filter Modules (see Fig.8), tug the Pull Chain(s) to release the Drain Down Filter plug(s). Allow the excess water to drain out of the chamber.
5. Use the skimmer pole to fit the Drain Down Filter plug back into the open port.
6. Once all floatables and oil have been removed, drop the vactor hose to the base of the sump. Vactor out the sediment and gross debris from the sump floor. Up to 0.3 yd<sup>3</sup> (0.2 m<sup>3</sup>) of sediment and 360 gallons (1,363 L) of water will be removed from a typical manhole Up-Flo® Filter during this process.
7. Retract the vactor hose from the vessel.
8. Inspect the Angled Screens for blockages and ragging. If present, remove the obstruction or ragging materials from the surface using a hose or other screen-cleaning device.
9. On the Maintenance Log provided by Hydro International, record the date, unit location, estimated volume of floatables, oils, and gross debris removed, and the depth of sediment measured. Note any apparent irregularities such as damaged components or blockages.
10. Securely replace the grate or lid. Remove safety equipment.
11. Dispose of sediment and gross debris following local regulations.
12. Dispose of oil and sump water at a licensed water treatment facility or following local regulations.
13. Contact Hydro International at (800) 848-2706 to discuss any irregularities noted during cleanup.

These activities include replacement of the Media Packs and Drain Down Filter.

Unless the Up-Flo® Filter has been installed as a very shallow unit, it is necessary to have an OSHA-confined space entry trained person enter the vessel to replace Media Packs.

The access port located at the top of the manhole or vault provides access to the Up-Flo® vessel for maintenance personnel to enter the vessel and remove and replace Media Packs. The same access would be used for maintenance personnel working from the surface to net or skim debris and floatables or to vactor out sediment, oil, and water. Unless the Up-Flo® Filter has been installed in a very shallow configuration, it is necessary to have personnel with OSHA Confined Space Entry training performing the maintenance that occurs inside the vessel.

**SCHEDULING**

- Call Hydro International to order replacement Media Packs and Drain Down Filter prior to scheduling maintenance.
- Because Media Pack replacement requires entry into the Up-Flo® chamber, maintenance events should be scheduled during dry weather.
- Media Pack replacement should occur immediately after a contaminated spill in the contributing drainage area.

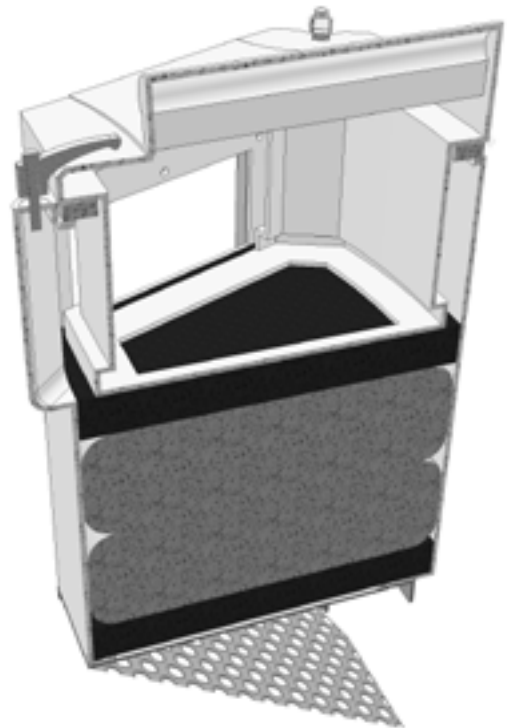


Fig.8 Cutaway view of the Filter Module

**MAINTENANCE ACTIVITIES REQUIRING MAN ENTRY**

### Recommended Equipment

- Safety Equipment (traffic cones, etc.)
- Crow bar to remove grate or lid
- Pole with skimmer or net (if floatables removal is not to be done with vactor hose)
- Sediment probe (such as a Sludge-Judge®)
- Vactor truck (flexible hose preferred)
- OSHA Confined Space Entry Equipment
- Up-Flo® Filter Replacement Media Packs (available from Hydro International)
- Hydro International Up-Flo® Filter Maintenance Log
- Screwdriver (flat head)
- Replacement Drain Down Filter components supplied by Hydro International

### Man Entry Required: Media Pack and Drain Down Filter

1. Follow Floatables and Sump Cleanout Procedures, 1 – 13.
2. Following OSHA Confined Space Entry procedures, enter the

Up-Flo® Filter Chamber.

3. Open the Filter Module by turning the three cam latches on the front and sides of the module. Remove the lid **1** to gain access to the Media Pack (Fig.9).
4. Remove and discard the spent Media Pack. The Media Pack contents include:
  - A top layer of **A** Flow Distributing Sheets
  - Two (2) Media Bags **B** equipped with nylon handles.
  - A bottom layer of **A** Flow Distributing Media.
5. Insert a new Media Pack, supplied by Hydro International.
  - First, insert a bottom layer of green Flow Distributing Media. Be sure that the media sits snugly and level at the bottom of the Filter Module.
  - Next, insert the first of two (2) replacement Media Bags. Smooth the bag out with your hands to make sure that the bag extends snugly to the walls and corners of the Filter Module.
  - Insert the second Media Bag, following the same procedure.
  - Insert the top layer of green Flow Distributing Media.

1. Filter Module Cover and Media Restraint

2. Replaceable Media Pack:

- a) Flow distribution sheets
- b) Filter Media Bags

3. Cam Latch

4. Conveyance Channel

5. Filter Module

6. Support Bracket / Angled Screen

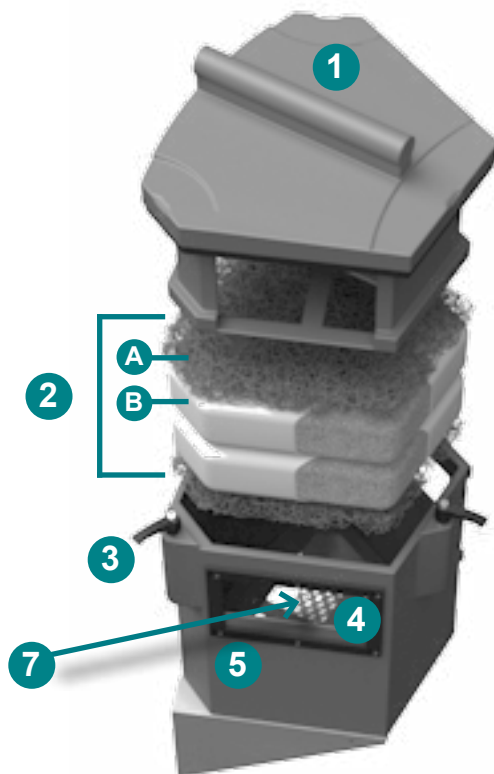


Fig.9 The Filter Module houses the Media Restraint and the Media Pack.

Be sure that the piece fits snugly against the walls and corners of the Filter Module.

- Put the lid on and secure the three latches. Check to make sure that the latches are closed properly.

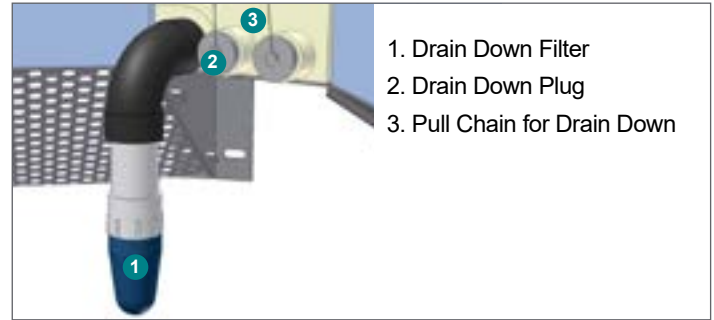
6. Use a screwdriver to unscrew the Drain Down Filter from the face of the Outlet Module (see Fig.10). **DO NOT DISCARD THIS PIECE.**

7. Install new Drain Down Filter supplied by Hydro International.

8. Exit the Up-Flo® Filter chamber and securely replace the grate \_\_\_ or lid.

9. On the Maintenance Log provided by Hydro International, record the date, unit location, estimated volume of floatables, oil and gross debris removed, and the depth of sediment measured. Note the number of Media Packs replaced. Note any irregularities such as damaged components or blockages.

**Fig.10** The Drain Down Filter.



10. Remove safety equipment.

11. Dispose of spent media packs at your local landfill, following local regulations.

12. Return the spent Drain Down Filter to Hydro International.

13. Contact Hydro International to discuss any irregularities noted during annual maintenance.

### Solids Disposal

Sediment, floatables, gross debris, and spent Media Bags can generally be disposed of at the local landfill in accordance with local regulations. The toxicity of the residues captured will depend on the activities in the contributing drainage area, and testing of the residues may be required if they are considered potentially hazardous.

Sump water can generally be disposed of at a licensed water treatment facility but the local sewer authority should be contacted for permission prior to discharging the liquid. Significant accumulations of oil removed separately from sump water should be transported to a licensed hazardous waste treatment facility for treatment or disposal. **In all cases, local regulators should be contacted about disposal requirements.**

## MAINTENANCE AT A GLANCE

| Activity                      | Frequency                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inspection                    | <ul style="list-style-type: none"> <li>- Regularly during first year of installation</li> <li>- Every 6 months after the first year of installation</li> </ul>                                                                                                                                                                                                                                     |
| Floatables/Oils Removal       | <ul style="list-style-type: none"> <li>- Twice per year or as needed</li> <li>- Following a contaminated spill in the drainage area</li> </ul>                                                                                                                                                                                                                                                     |
| Sediment Removal              | <ul style="list-style-type: none"> <li>- Every six to 12 months, depending on the Up-Flo® Filter Configuration</li> <li>- The maximum allowable sediment depth in any Up-Flo Filter configuration is 16 inches (41 cm)</li> <li>- Following a contaminated spill in the drainage area</li> </ul>                                                                                                   |
| Media Pack Replacement        | <ul style="list-style-type: none"> <li>- Once per year</li> <li>- Replacement is required anytime inspection reveals that the high-water level indicator has been activated after two consecutive storms and the subsequent weighing of the Media Bags shows a wet weight greater than 40 lbs</li> <li>- Following a contaminated spill in the drainage area</li> </ul>                            |
| Drain Down Filter Replacement | <ul style="list-style-type: none"> <li>- Once per year with Media Pack replacement</li> <li>- Replacement is required anytime inspection reveals that the water level inside the vessel has not reached a level equal with the base of the Filter Modules approximately 36 hours after a 1-inch (2.5 cm) rainfall</li> <li>- As needed, in the event of continuous base flow conditions</li> </ul> |

## UP-FLO® FILTER INSTALLATION LOG



|                                                                        |                         |
|------------------------------------------------------------------------|-------------------------|
| <b>SITE REFERENCE NAME OR NUMBER FOR THIS UP-FLO® FILTER LOCATION:</b> |                         |
| <b>SITE NAME:</b>                                                      |                         |
| <b>SITE LOCATION:</b>                                                  |                         |
| <b>OWNER:</b>                                                          | <b>SITE CONTRACTOR:</b> |
| <b>CONTACT NAME:</b>                                                   | <b>CONTACT NAME:</b>    |
| <b>COMPANY NAME:</b>                                                   | <b>COMPANY NAME:</b>    |
| <b>ADDRESS:</b>                                                        | <b>ADDRESS:</b>         |
| <b>TELEPHONE:</b>                                                      | <b>TELEPHONE:</b>       |
| <b>FAX:</b>                                                            | <b>FAX:</b>             |

INSTALLATION DATE:     /     /

CONFIGURATION (CIRCLE ONE):     MANHOLE     VAULT SYSTEM

TOTAL NUMBER OF UP-FLO® FILTER MODULES: \_\_\_\_\_



## UP-FLO® FILTER INSPECTION LOG

Site Name: \_\_\_\_\_ Owner Change since last inspection? Y N

Location: \_\_\_\_\_

Owner Name: \_\_\_\_\_

Address: \_\_\_\_\_ Phone Number: \_\_\_\_\_

Site Status: \_\_\_\_\_

Date: \_\_\_\_\_ Time: \_\_\_\_\_ Site conditions\*: \_\_\_\_\_  
 \*(Stable, Under Construction, Needing Maintenance, etc.)

Inspection Frequency Key: A=annual; M=monthly; S=after major storms

| Inspection Items                                                                         | Inspection Frequency | Inspected? (Yes/No) | Maintenance Needed? (Yes/No) | Comments/Description |
|------------------------------------------------------------------------------------------|----------------------|---------------------|------------------------------|----------------------|
| <b>Debris Removal</b>                                                                    |                      |                     |                              |                      |
| Adjacent area free of debris?                                                            | M                    |                     |                              |                      |
| Inlets and Outlets free of debris?                                                       | M                    |                     |                              |                      |
| Facility (internally) free of debris?                                                    | M                    |                     |                              |                      |
| <b>Vegetation</b>                                                                        |                      |                     |                              |                      |
| Surrounding area fully stabilized? (no evidence of eroding material into Up-Flo® Filter) | A                    |                     |                              |                      |
| Grass mowed?                                                                             | M                    |                     |                              |                      |
| <b>Water retention where required</b>                                                    |                      |                     |                              |                      |
| Water holding chamber(s) at normal pool?                                                 | A                    |                     |                              |                      |
| Evidence of erosion?                                                                     | A                    |                     |                              |                      |
| <b>Sediment Deposition</b>                                                               |                      |                     |                              |                      |
| Filtration Chamber free of sediments?                                                    | A                    |                     |                              |                      |
| Sedimentation sump not more than 50% full?                                               | A                    |                     |                              |                      |
| <b>Structural Components</b>                                                             |                      |                     |                              |                      |
| Any evidence of structural deterioration?                                                | A                    |                     |                              |                      |
| Grates in good condition?                                                                | A                    |                     |                              |                      |
| Spalling or cracking of structural parts?                                                | A                    |                     |                              |                      |
| Outlet/Overflow Spillway                                                                 | A                    |                     |                              |                      |
| <b>Other</b>                                                                             |                      |                     |                              |                      |
| Noticeable odors?                                                                        | A                    |                     |                              |                      |
| Any evidence of filter(s) clogging?                                                      | M                    |                     |                              |                      |
| Evidence of flow bypassing facility?                                                     | A                    |                     |                              |                      |



Inspector Comments: \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

Overall Condition of Up-Flo® Filter\*\*: ☐ Acceptable ☐ Unacceptable

*\*\*"Acceptable" would mean properly functioning; "unacceptable" would mean damaged or required further maintenance.*

If any of the above Inspection Items are checked "Yes" for "Maintenance Needed", list Maintenance actions and their completion dates below or on the Maintenance Log provided on page 15 of the Up-Flo® Filter Operation & Maintenance Manual:

| Maintenance Action Needed | Due Date |
|---------------------------|----------|
|                           |          |
|                           |          |
|                           |          |
|                           |          |
|                           |          |

The next routine inspection is schedule for approximately: (date) \_\_\_\_\_

Inspected by: (signature) \_\_\_\_\_

Inspected by: (printed) \_\_\_\_\_



## UP-FLO® FILTER MAINTENANCE LOG

Site Name: \_\_\_\_\_ Owner Change since last inspection? Y N

Location: \_\_\_\_\_

Owner Name: \_\_\_\_\_

Address: \_\_\_\_\_ Phone Number: \_\_\_\_\_

Site Status: \_\_\_\_\_

Date: \_\_\_\_\_ Time: \_\_\_\_\_ Site conditions: \_\_\_\_\_  
*\*(Stable, Under Construction, Needing Maintenance, etc.)*

Estimated volume of oil/floatable trash removed: \_\_\_\_\_

Sediment depth measured in sump prior to removal: \_\_\_\_\_

Number of Filter Modules fitted with new media packs: \_\_\_\_\_

Inspector Comments: \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

Overall Condition of Up-Flo® Filter: ☐ Acceptable ☐ Unacceptable

*\*\*"Acceptable" would mean properly functioning; "unacceptable" would mean damaged or required further maintenance.*

Maintained by: (signature) \_\_\_\_\_

Maintained by: (printed) \_\_\_\_\_

# DO IT RIGHT THE FIRST TIME

LEARN MORE AT [HYDRO-INT.COM/SERVICE](http://HYDRO-INT.COM/SERVICE)



CALL 1 (888) 382-7808 TO SCHEDULE AN INSPECTION

## Stormwater Solutions

94 Hutchins Drive  
Portland, ME 04102

Tel: (207) 756-6200  
Fax: (207) 756-6212  
[stormwaterinquiry@hydro-int.com](mailto:stormwaterinquiry@hydro-int.com)

[www.hydro-int.com](http://www.hydro-int.com)

Turning Water Around...®

UFF\_OMM\_D1608



January 5, 2021

Travis Owen  
Southeast Culvert  
1094 Bankhead Highway  
Winder, GA 30680

RE: Proprietary Device Application for General Use/ Up-Flo® Filter system

Dear Mr. Owen:

This letter is to document the review of the Proprietary Device Committee and the Gwinnett County response concerning the application for the general use of an Up-Flo® Filter system proprietary device as designed in 2020, in accordance with the Unified Development Ordinance (UDO) and the GCSMM, for use in unincorporated Gwinnett County. The Up-Flo® Filter system is expected to provide the required 80% TSS removal from the discharged Water Quality Volume. After review and discussion, the GCSMM Committee recommends approval of the Up-Flo® Filter system proprietary device as designed in 2020 for general use in unincorporated Gwinnett County, under the following conditions.

The following requirements apply to this construction project:

1. The design, installation, and operation shall adhere to all manufacturing recommendations, including:
  - a. Discharge to the Up-Flo® Filter system shall be limited to stormwater runoff.
  - b. No greater than 250 lbs. annual sediment load per module.
  - c. Maximum of 0.30 impervious acres treated per filter module.
  - d. Maximum treatment flowrate of 20 gallons/minute (0.045 cfs) per filter module. Up-Flo® Filter structure must be deep enough to allow a 30" operating head inside the structure (as measured from the outlet invert) and 30" minimum operating head shall be provided, unless otherwise calculated and approved in writing by the manufacturer.
  - e. Sufficient bypass capacity shall be provided for all storms greater than the Water Quality Storm.
  - f. The minimum difference in elevation between the inflow and the outflow pipes of the Up-Flo® Filter system shall be 0.8 ft. (9.48 in.) unless authorized by and documented to Gwinnett County by the manufacturer, in writing.
  - g. The Up-Flo® Filter structure must be accessible by a vacuum truck.

- h. There must be enough space around the Up-Flo® Filter structure to be able to perform an OSHA approved confined space entry.
2. The Up-Flo® Filter system may only be installed in projects with the following site conditions:
  - a. Commercial sites, not residential or multi-family sites;
  - b. Limited room for larger footprint BMPs;
  - c. Sites on which infiltration BMPs would be inappropriate (e.g. fuel sales stations); and/or soil has insufficient permeability for infiltration BMPs; and/or
  - d. As justified by the engineer of record.
3. The Up-Flo® Filter may be installed upstream or downstream of a detention pond, underground detention, or other type of BMP used to meet the Channel Protection or Peak Flow Stormwater Performance Standards, as long as the following conditions are met:
  - a. The manufacturer of the Up-Flo® Filter system proprietary device determines the sizing of this unit and provides the sizing calculations and authorizes such use in writing. These calculations must follow the GCSMM and the methodology outlined in the Site Development Review Tool Spreadsheet.
  - b. The design engineer and the manufacturer must ensure the flows, including volume and head, directed to the Up-Flo® Filter Unit do not adversely impact the functionality of the unit.
  - c. The Up-Flo® Filter system may not be nested within another BMP such as a detention pond, underground detention, or other type of BMP. It is to be installed as a standalone BMP or configured in a treatment train concept.
  - d. Any BMP configured with the Up-Flo® Filter system must be able to empty as designed, as defined in the GCSMM.
4. During the first year after installation the Up-Flo® Filter structure must be inspected at least once after a one-inch rain event. During operation, periodic inspections must be on a quarterly basis.
  - a. During operation of the Up-Flo® Filter structure, do not allow oil to accumulate to the point where it has formed a measurable thickness on the surface of the water.
  - b. During operation of the Up-Flo® Filter structure, do not allow sediment to block the inlet to the drain down filter.
5. After the first year of installation, the owner must submit to the Department of Water Resources (DWR), MS4 Compliance Group, the following information:
  - a. Maintenance records with frequency based on the first year's operation. Include all supporting data as specified on page 5 of the Operations and Maintenance Manual.
  - b. A blank inspection form
  - c. A blank maintenance form

6. A Maintenance Agreement is required as specified in the GCSMM, and it must require the following:
  - a. Periodic inspections must be on a quarterly basis.
  - b. Maintenance must be performed on the frequency defined in the first year's operations. A new maintenance frequency may be established using new supporting data as specified on page 5 of the Operations and Maintenance Manual.
  - c. items 4(a), 4(b).
7. Annually submit inspection reports and maintenance reports, using the forms noted in item #5, to the Department of Water Resources (DWR), MS4 Compliance Group.
8. If a project includes an upgraded Up-Flo® Filter system, redesigned or reconfigured so as to be substantially different than the General Application submitted in 2020, a new application must be submitted.

This approval is also dependent on the full plan review and approval of the project's stormwater management plan by the Department of Planning and Development.

This decision may be appealed to the Stormwater Authority. Appeals must be submitted in writing and provide clear, concise reasoning as to why the appeal should be granted. Please submit such documentation to the Director of the Department of Water Resources in a timely fashion.

We appreciate the opportunity to review this proprietary device. Should you have any questions regarding this approval or should you require any further information, please contact this office at 678-376-6923 or email me at [Tyler.Richards@gwinnettcountry.com](mailto:Tyler.Richards@gwinnettcountry.com).

Sincerely,



Tyler Richards  
Department Director  
Department of Water Resources  
684 Winder Highway  
Lawrenceville, GA 30045

LIDL #1517

4990 JIMMY CARTER BLVD, NORCROSS 30093

GWINNETT CO, GA

Feature to be updated  
based on revised civil  
design plans that will be  
issued at a later date.

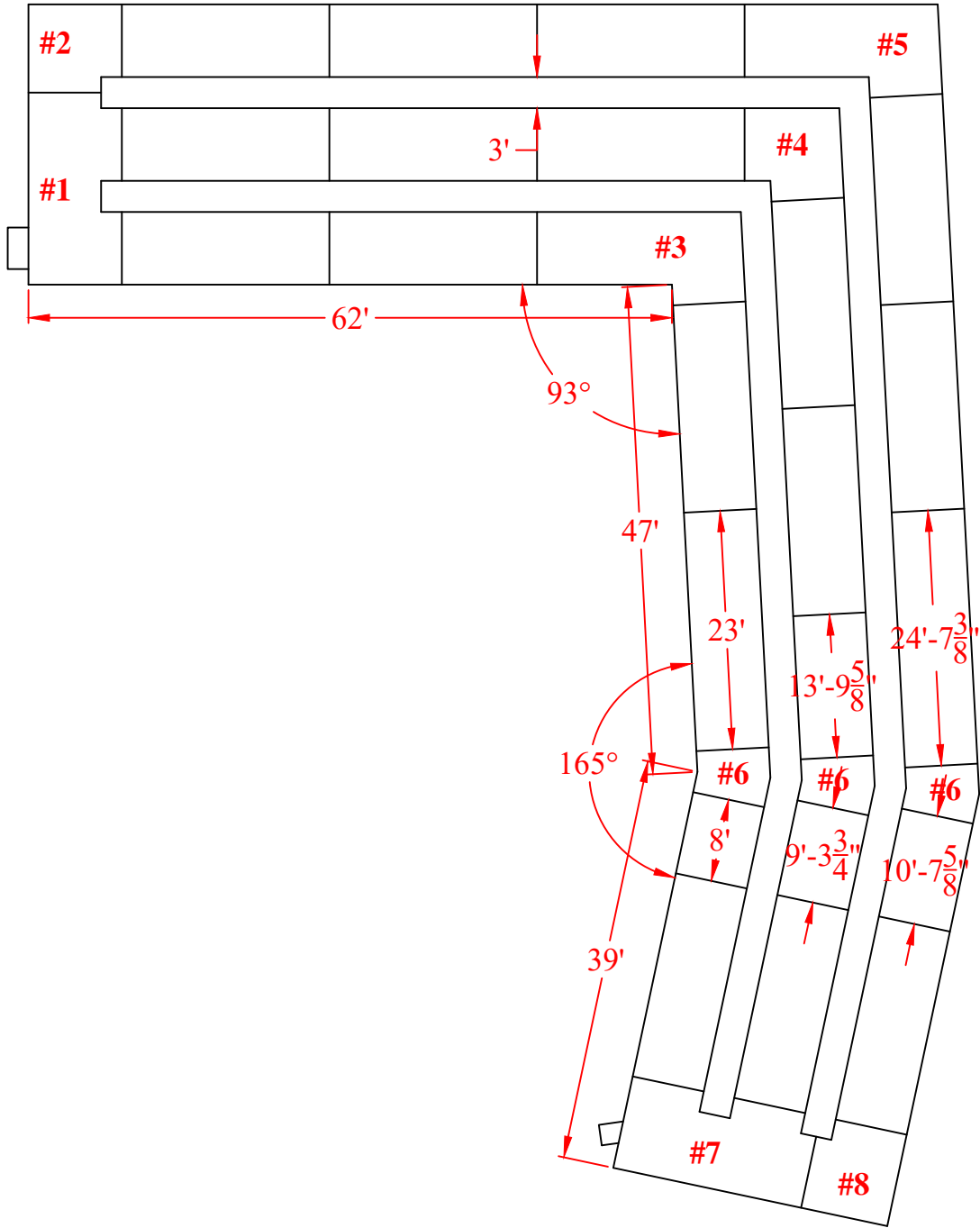
Submittal Drawings Sheets T101-D103 for approval:

Approved By:\_\_\_\_\_

Date:\_\_\_\_\_

|     |             |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                        |            |        |        |              |                 |
|-----|-------------|-----|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------|--------|--------------|-----------------|
| REV | DESCRIPTION | BY  | DATE | PROPRIETARY & CONFIDENTIAL: THIS DRAWING IS THE SOLE PROPERTY OF SOUTHEAST CULVERT, INC. ("SEC") AND CONTAINS PROPRIETARY INFORMATION APPLICABLE TO THIS PROJECT ONLY. ANY DIFFERENCES BETWEEN THE DESIGN INFORMATION, THE GENERAL NOTES AND FIELD CONDITIONS MUST BE REPORTED TO SEC IMMEDIATELY FOR REEVALUATION. SEC IS NOT LIABLE FOR INCOMPLETE OR INACCURATE DRAWINGS, DRAWINGS THAT ARE UPDATED WITHOUT SEC PERMISSION, OR CHANGES IN FIELD CONDITIONS. ANY REPRODUCTION OR MODIFICATION WITHOUT WRITTEN PERMISSION IS STRICTLY PROHIBITED. | <div><div><div><div><div></div></div><div></div></div><div>SOUTHEAST<br/>CULVERT</div></div><div>1094 Bankhead Hwy, Winder, GA 30680<br/>PH: 770-868-5599   Fax: 770-868-5735<br/>www.southeastculvert.com</div></div> | LIDL #1517 | COVER  |        | DESIGNED BY: | PROJECT NUMBER: |
| ### | ###         | ### | ###  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                        |            |        |        | DRAWN BY:    | SHEET NO. OF    |
|     |             |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                        |            |        |        | MHM          |                 |
|     |             |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                        |            | DATE:  | SCALE: | CHECKED BY:  |                 |
|     |             |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                        |            | 4/1/26 | NA     |              | 1 / 12          |
|     |             |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                        |            |        |        |              |                 |

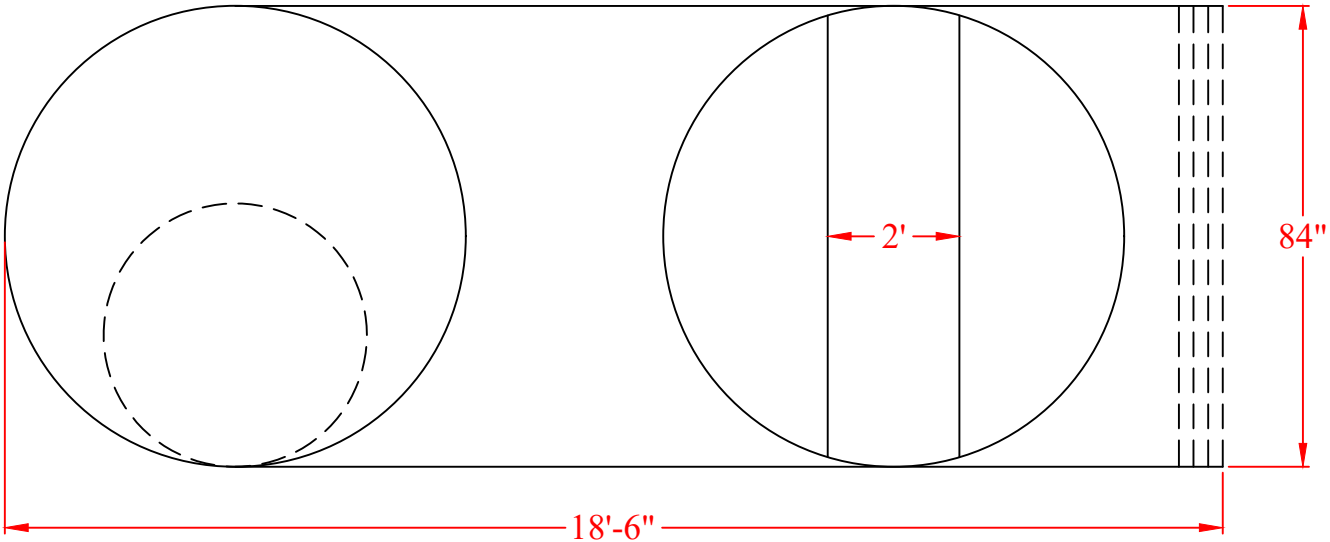
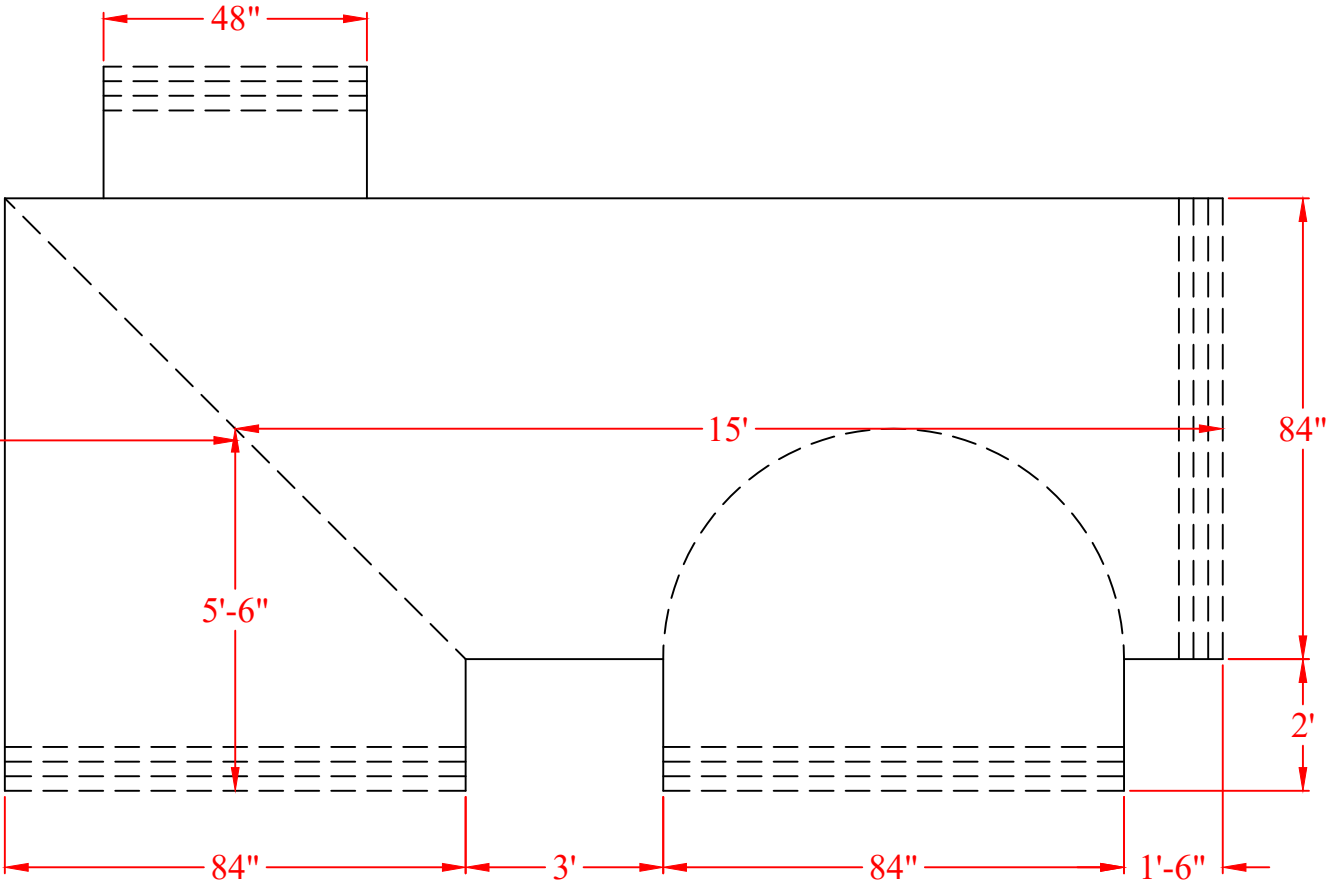
- PROJECT SPECIFIC NOTES:
1. SOUTHEAST CULVERT FABRICATION DRAWINGS ARE COPYRIGHTED. THEY MAY NOT BE REPRODUCED, COPIED, OR ISSUED TO A 3RD PARTY WITHOUT WRITTEN PERMISSION OF SOUTHEAST CULVERT.
  2. ALL DIMENSIONS ARE SUBJECT TO MANUFACTURING TOLERANCES PER REFERENCED SPECIFICATIONS, INDUSTRY STANDARDS AND INSTALLATION TOLERANCES.
  3. ALL DIMENSIONS ARE IN FEET AND INCHES UNLESS OTHERWISE NOTED
  4. ALL PIPE DIMENSIONS ARE MEASURED TO THE NEUTRAL AXIS OF THE CORRUGATION PROFILE UNLESS OTHERWISE NOTED.
  5. ALL ELEVATIONS ARE IN FEET UNLESS OTHERWISE NOTED
  6. ALL CORRUGATED METAL PIPE AND CORRUGATED METAL PIPE STUBS SHALL BE ALUMINIZED STEEL TYPE 2 per AASHTO M 274.  
  
PIPE SHALL BE MANUFACTURED PER AASHTO M 36 AND SHALL HAVE REROLLED ENDS IF CONNECTING TO ANOTHER CORRUGATED METAL PIPE UNLESS OTHERWISE NOTED.
  7. 84" PIPE SHALL BE 5" X 1" CORRUGATION 14 GAGE MINIMUM.
  8. ALL ACCESS RISERS SHOWN ON THE PLANS ARE FABRICATED APPROXIMATELY 1.5' SHORT OF FINISHED GRADE - THIS SHOULD PROVIDE ROOM FOR A REINFORCED CONCRETE CAP (BY OTHERS)
  9. ALL FABRICATED FITTINGS SHALL MARKED ACCORDINGLY. UNMARKED PIPE LENGTHS ARE 20' IN LENGTH



**\*84", 14 GAGE, 5X1, AT2.**  
**\*UNLABELED PIECES ARE 20' LENGTHS.**  
**\*PIPE TO BE ALUMINIZED STEEL TYPE 2 PER AASHTO M274.**  
**\*SYSTEM TO BE INSTALLED PER AASHTO SECTION 26.**

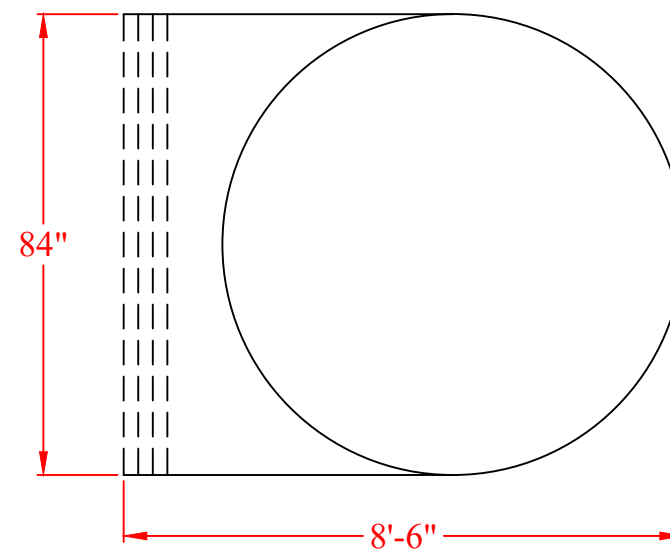
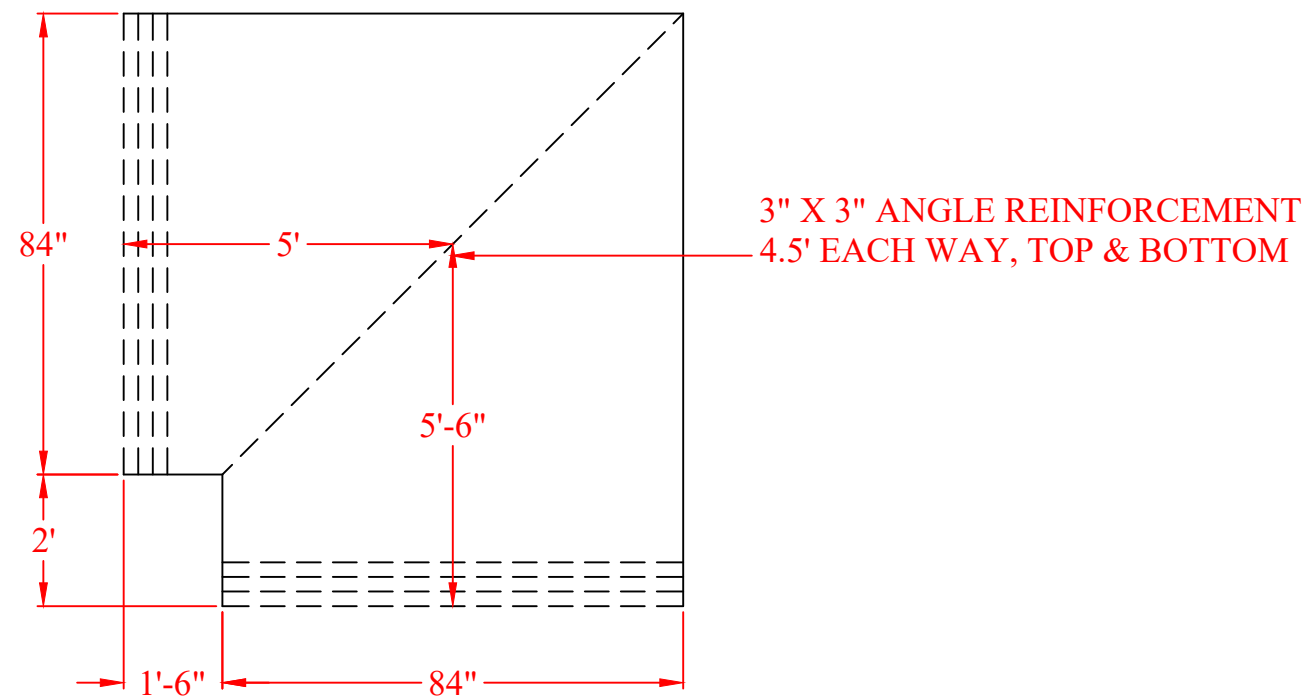
| REV  | DESCRIPTION | BY   | DATE | PROPRIETARY & CONFIDENTIAL: THIS DRAWING IS THE SOLE PROPERTY OF SOUTHEAST CULVERT, INC. ("SEC") AND CONTAINS PROPRIETARY INFORMATION APPLICABLE TO THIS PROJECT ONLY. ANY DIFFERENCES BETWEEN THE DESIGN INFORMATION, THE GENERAL NOTES AND FIELD CONDITIONS MUST BE REPORTED TO SEC IMMEDIATELY FOR REEVALUATION. SEC IS NOT LIABLE FOR INCOMPLETE OR INACCURATE DRAWINGS, DRAWINGS THAT ARE UPDATED WITHOUT SEC PERMISSION, OR CHANGES IN FIELD CONDITIONS. ANY REPRODUCTION OR MODIFICATION WITHOUT WRITTEN PERMISSION IS STRICTLY PROHIBITED. | <div><div><div><div><div><div><span></span></div></div></div><div><div><span></span></div></div></div><div>SOUTHEAST<br/>CULVERT</div></div><div>1094 Bankhead Hwy, Winder, GA 30680<br/>PH: 770-868-5599   Fax: 770-868-5735<br/>www.southeastculvert.com</div></div> | LIDL #1517 |  | SYSTEM LAYOUT |        | DESIGNED BY: | PROJECT NUMBER: |
|------|-------------|------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--|---------------|--------|--------------|-----------------|
| #### | ####        | #### | #### |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                        |            |  |               |        | DRAWN BY:    | SHEET NO. OF    |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                        |            |  |               |        | MHM          |                 |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                        |            |  |               |        | CHECKED BY:  | 2 / 12          |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                        |            |  |               |        |              |                 |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                        |            |  |               |        |              |                 |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                        |            |  |               |        |              |                 |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                        |            |  | DATE:         | SCALE: |              |                 |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                        |            |  | 4/1/26        | NA     |              |                 |

3" X 3" ANGLE REINFORCEMENT  
4.5' EACH WAY, TOP & BOTTOM



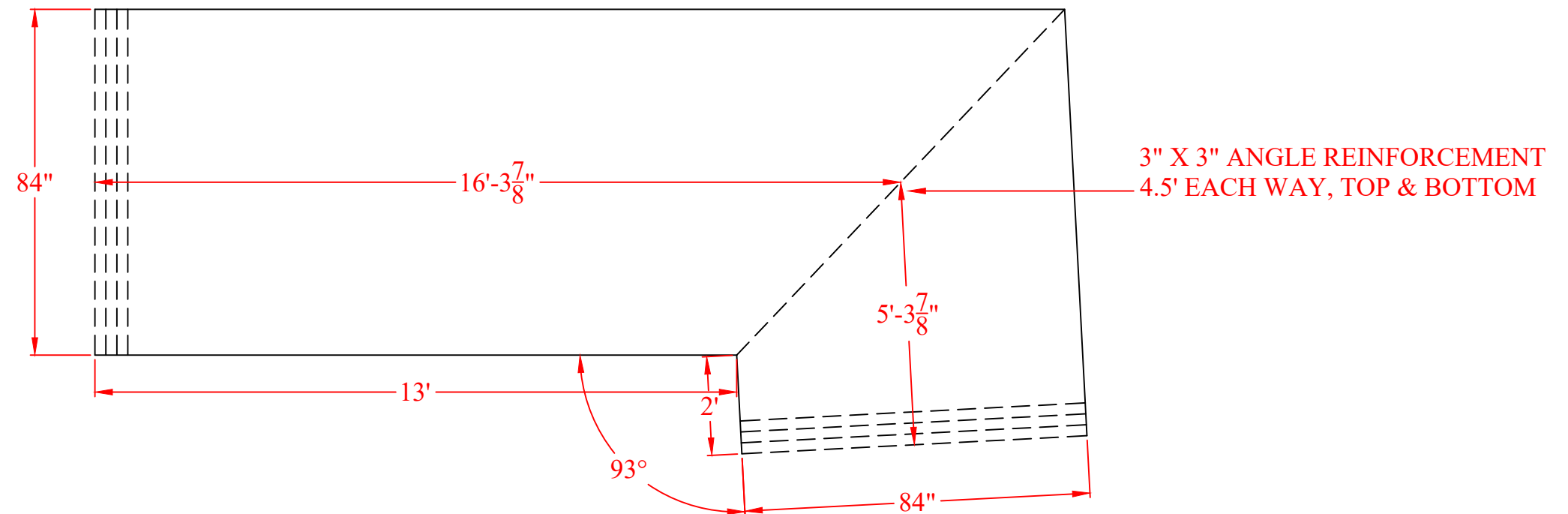
\*14 GAGE, 5X1, AT2.

| REV | DESCRIPTION | BY  | DATE | PROPRIETARY & CONFIDENTIAL: THIS DRAWING IS THE SOLE PROPERTY OF SOUTHEAST CULVERT, INC. ("SEC") AND CONTAINS PROPRIETARY INFORMATION APPLICABLE TO THIS PROJECT ONLY. ANY DIFFERENCES BETWEEN THE DESIGN INFORMATION, THE GENERAL NOTES AND FIELD CONDITIONS MUST BE REPORTED TO SEC IMMEDIATELY FOR REEVALUATION. SEC IS NOT LIABLE FOR INCOMPLETE OR INACCURATE DRAWINGS, DRAWINGS THAT ARE UPDATED WITHOUT SEC PERMISSION, OR CHANGES IN FIELD CONDITIONS. ANY REPRODUCTION OR MODIFICATION WITHOUT WRITTEN PERMISSION IS STRICTLY PROHIBITED. |  |  | SOUTHEAST CULVERT |  | LIDL #1517 |  | MANIFOLD #1 |  | DESIGNED BY: | PROJECT NUMBER: |
|-----|-------------|-----|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|-------------------|--|------------|--|-------------|--|--------------|-----------------|
| ### | ###         | ### | ###  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |                   |  |            |  |             |  | DRAWN BY:    | SHEET NO. OF    |
|     |             |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |                   |  |            |  |             |  | MHM          |                 |
|     |             |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |                   |  |            |  |             |  | CHECKED BY:  | 3 / 12          |
|     |             |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |                   |  |            |  |             |  | DATE: 4/1/26 | SCALE: NA       |
|     |             |     |      | 1094 Bankhead Hwy, Winder, GA 30680<br>PH: 770-868-5599   Fax: 770-868-5735<br>www.southeastculvert.com                                                                                                                                                                                                                                                                                                                                                                                                                                            |  |  |                   |  |            |  |             |  |              |                 |

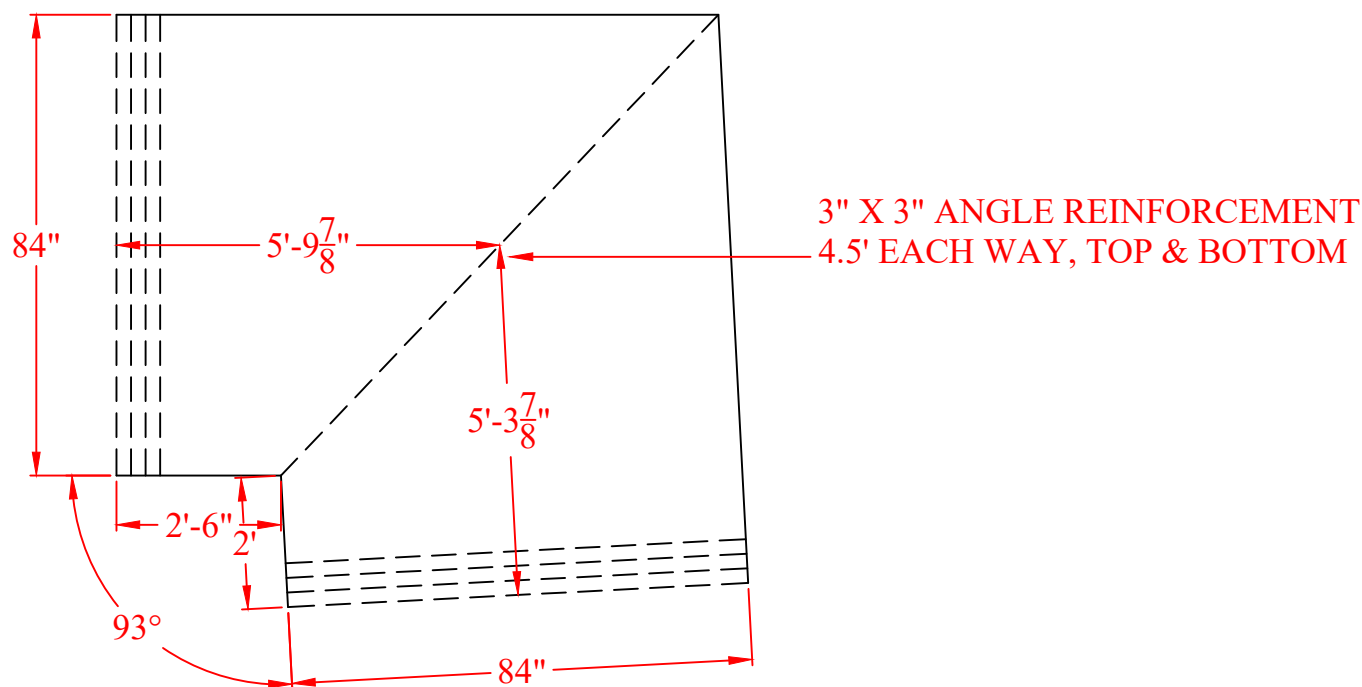


**\*14 GAGE, 5X1, AT2.**

|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                |            |             |  |                  |                 |              |             |
|------|-------------|------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|--|------------------|-----------------|--------------|-------------|
| REV  | DESCRIPTION | BY   | DATE | PROPRIETARY & CONFIDENTIAL: THIS DRAWING IS THE SOLE PROPERTY OF SOUTHEAST CULVERT, INC. ("SEC") AND CONTAINS PROPRIETARY INFORMATION APPLICABLE TO THIS PROJECT ONLY. ANY DIFFERENCES BETWEEN THE DESIGN INFORMATION, THE GENERAL NOTES AND FIELD CONDITIONS MUST BE REPORTED TO SEC IMMEDIATELY FOR REEVALUATION. SEC IS NOT LIABLE FOR INCOMPLETE OR INACCURATE DRAWINGS, DRAWINGS THAT ARE UPDATED WITHOUT SEC PERMISSION, OR CHANGES IN FIELD CONDITIONS. ANY REPRODUCTION OR MODIFICATION WITHOUT WRITTEN PERMISSION IS STRICTLY PROHIBITED. | <div> <b>SOUTHEAST CULVERT</b></div> <div>1094 Bankhead Hwy, Winder, GA 30680<br/>PH: 770-868-5599   Fax: 770-868-5735<br/>www.southeastculvert.com</div> | LIDL #1517 | MANIFOLD #2 |  | DESIGNED BY:     | PROJECT NUMBER: |              |             |
| #### | ####        | #### | #### |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                |            |             |  | DRAWN BY:<br>MHM | SHEET NO. OF    |              |             |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                |            |             |  |                  |                 |              | 4 / 12      |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                |            |             |  |                  |                 |              |             |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                |            |             |  |                  |                 |              |             |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                |            |             |  |                  | DATE:<br>4/1/26 | SCALE:<br>NA | CHECKED BY: |

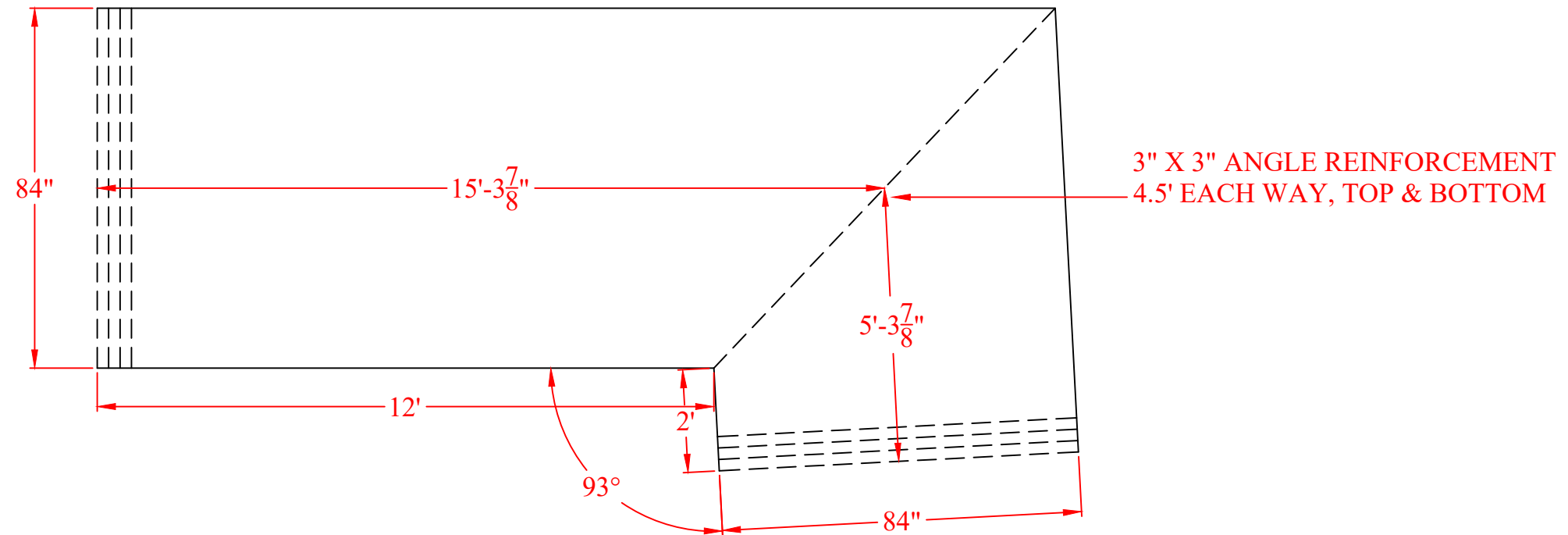


|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                |            |          |    |              |                 |        |             |
|------|-------------|------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|----|--------------|-----------------|--------|-------------|
| REV  | DESCRIPTION | BY   | DATE | PROPRIETARY & CONFIDENTIAL: THIS DRAWING IS THE SOLE PROPERTY OF SOUTHEAST CULVERT, INC. ("SEC") AND CONTAINS PROPRIETARY INFORMATION APPLICABLE TO THIS PROJECT ONLY. ANY DIFFERENCES BETWEEN THE DESIGN INFORMATION, THE GENERAL NOTES AND FIELD CONDITIONS MUST BE REPORTED TO SEC IMMEDIATELY FOR REEVALUATION. SEC IS NOT LIABLE FOR INCOMPLETE OR INACCURATE DRAWINGS, DRAWINGS THAT ARE UPDATED WITHOUT SEC PERMISSION, OR CHANGES IN FIELD CONDITIONS. ANY REPRODUCTION OR MODIFICATION WITHOUT WRITTEN PERMISSION IS STRICTLY PROHIBITED. | <div> <b>SOUTHEAST CULVERT</b></div> <div>1094 Bankhead Hwy, Winder, GA 30680<br/>PH: 770-868-5599   Fax: 770-868-5735<br/>www.southeastculvert.com</div> | LIDL #1517 | ELBOW #3 |    | DESIGNED BY: | PROJECT NUMBER: |        |             |
| #### | ####        | #### | #### |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                |            |          |    | DRAWN BY:    | SHEET NO. OF    |        |             |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                |            |          |    |              |                 |        |             |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                |            |          |    |              |                 |        |             |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                |            |          |    |              |                 |        |             |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                |            |          |    |              | DATE:           | SCALE: | CHECKED BY: |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                |            | 4/1/26   | NA |              | 5 / 12          |        |             |



**\*14 GAGE, 5X1, AT2.**

| REV | DESCRIPTION | BY  | DATE | <p>PROPRIETARY &amp; CONFIDENTIAL: THIS DRAWING IS THE SOLE PROPERTY OF SOUTHEAST CULVERT, INC. ("SEC") AND CONTAINS PROPRIETARY INFORMATION APPLICABLE TO THIS PROJECT ONLY. ANY DIFFERENCES BETWEEN THE DESIGN INFORMATION, THE GENERAL NOTES AND FIELD CONDITIONS MUST BE REPORTED TO SEC IMMEDIATELY FOR REEVALUATION. SEC IS NOT LIABLE FOR INCOMPLETE OR INACCURATE DRAWINGS, DRAWINGS THAT ARE UPDATED WITHOUT SEC PERMISSION, OR CHANGES IN FIELD CONDITIONS. ANY REPRODUCTION OR MODIFICATION WITHOUT WRITTEN PERMISSION IS STRICTLY PROHIBITED.</p> | <p><br/>1094 Bankhead Hwy, Winder, GA 30680<br/>PH: 770-868-5599   Fax: 770-868-5735<br/>www.southeastculvert.com</p> | <p>LIDL #1517</p> | <p>ELBOW #4</p> |        | DESIGNED BY: | PROJECT NUMBER: |
|-----|-------------|-----|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-----------------|--------|--------------|-----------------|
| ### | ###         | ### | ###  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                            |                   |                 |        | DRAWN BY:    | SHEET NO. OF    |
|     |             |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                            |                   |                 |        | MHM          |                 |
|     |             |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                            |                   | DATE:           | SCALE: | CHECKED BY:  |                 |
|     |             |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                            |                   | 4/1/26          | NA     |              |                 |
|     |             |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                            |                   |                 |        |              |                 |

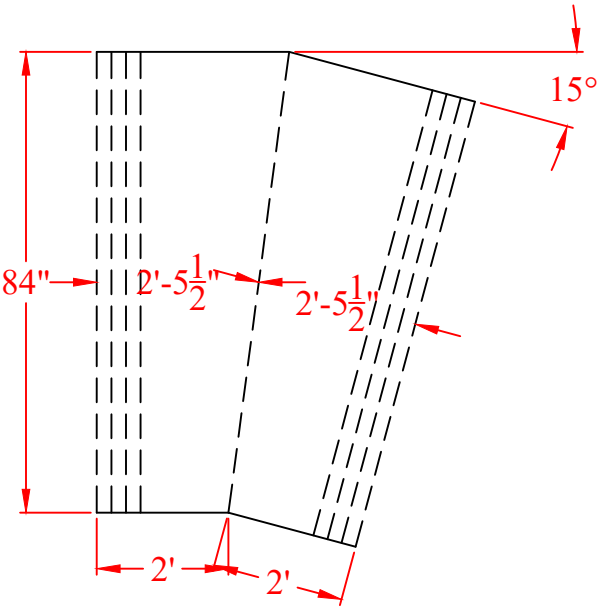


**\*14 GAGE, 5X1, AT2.**

|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                            |            |          |  |              |                 |        |             |        |
|------|-------------|------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|--|--------------|-----------------|--------|-------------|--------|
| REV  | DESCRIPTION | BY   | DATE | PROPRIETARY & CONFIDENTIAL: THIS DRAWING IS THE SOLE PROPERTY OF SOUTHEAST CULVERT, INC. ("SEC") AND CONTAINS PROPRIETARY INFORMATION APPLICABLE TO THIS PROJECT ONLY. ANY DIFFERENCES BETWEEN THE DESIGN INFORMATION, THE GENERAL NOTES AND FIELD CONDITIONS MUST BE REPORTED TO SEC IMMEDIATELY FOR REEVALUATION. SEC IS NOT LIABLE FOR INCOMPLETE OR INACCURATE DRAWINGS, DRAWINGS THAT ARE UPDATED WITHOUT SEC PERMISSION, OR CHANGES IN FIELD CONDITIONS. ANY REPRODUCTION OR MODIFICATION WITHOUT WRITTEN PERMISSION IS STRICTLY PROHIBITED. | <div>SOUTHEAST<br/>CULVERT</div> <div>1094 Bankhead Hwy, Winder, GA 30680<br/>PH: 770-868-5599   Fax: 770-868-5735<br/>www.southeastculvert.com</div> | LIDL #1517 | ELBOW #5 |  | DESIGNED BY: | PROJECT NUMBER: |        |             |        |
| #### | ####        | #### | #### |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                            |            |          |  | DRAWN BY:    | SHEET NO. OF    |        |             |        |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                            |            |          |  | MHM          |                 |        |             |        |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                            |            |          |  |              | DATE:           | SCALE: | CHECKED BY: | 7 / 12 |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                            |            |          |  | 4/1/26       | NA              |        |             |        |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                            |            |          |  |              |                 |        |             |        |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                            |            |          |  |              |                 |        |             |        |

**\*\*3 OF THESE\*\***

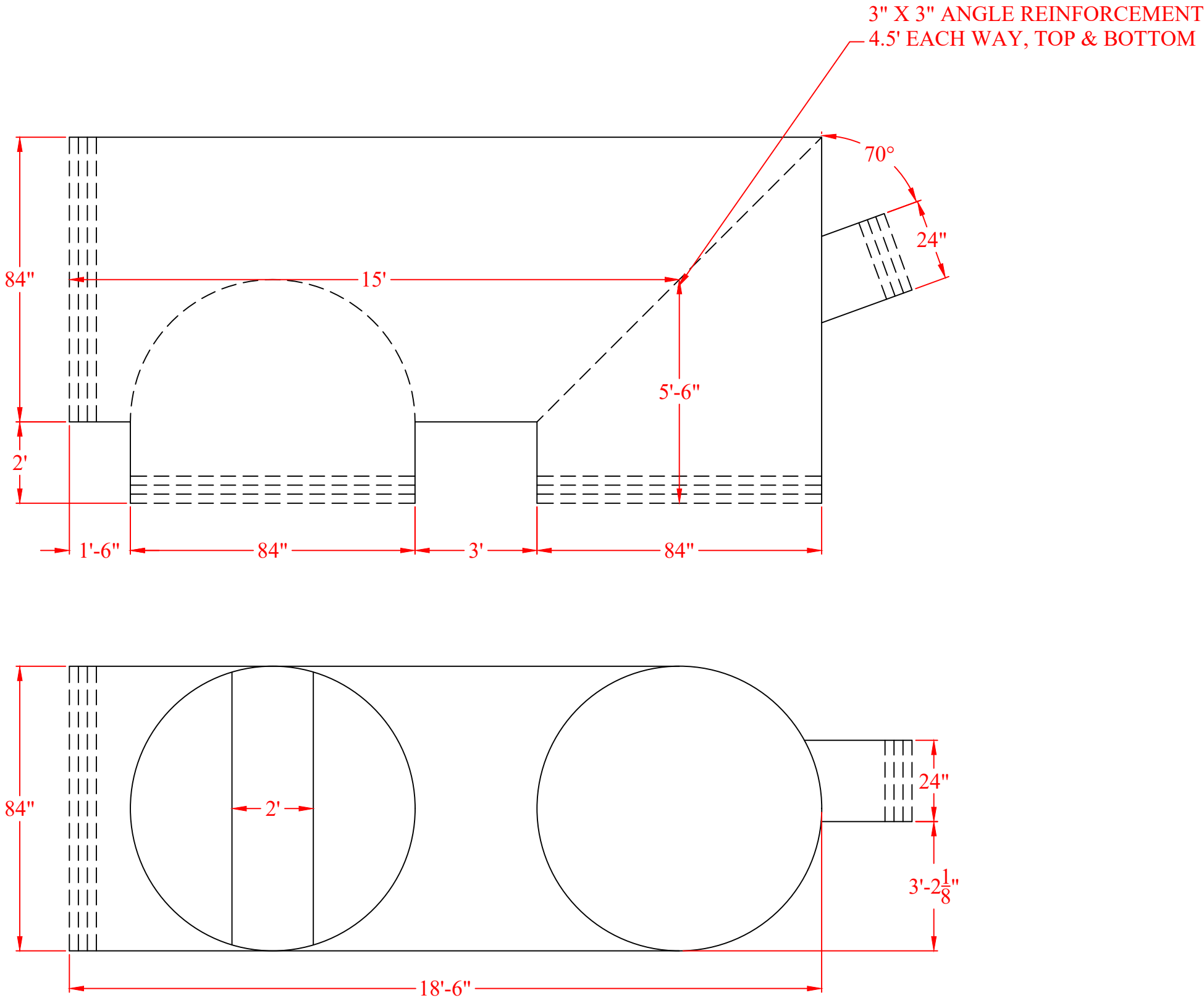
**\*\*3 OF THESE\*\***



**\*14 GAGE, 5X1, AT2.**

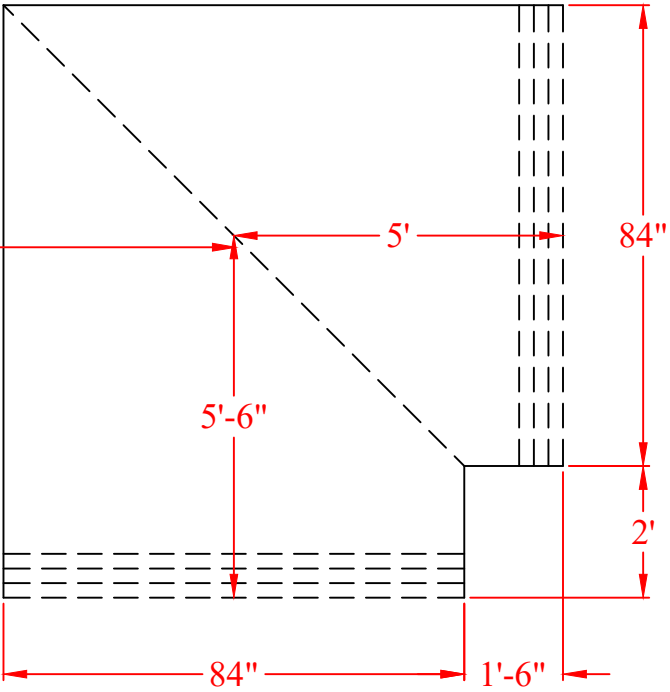
| REV  | DESCRIPTION | BY   | DATE | PROPRIETARY & CONFIDENTIAL: THIS DRAWING IS THE SOLE PROPERTY OF SOUTHEAST CULVERT, INC. ("SEC") AND CONTAINS PROPRIETARY INFORMATION APPLICABLE TO THIS PROJECT ONLY. ANY DIFFERENCES BETWEEN THE DESIGN INFORMATION, THE GENERAL NOTES AND FIELD CONDITIONS MUST BE REPORTED TO SEC IMMEDIATELY FOR REEVALUATION. SEC IS NOT LIABLE FOR INCOMPLETE OR INACCURATE DRAWINGS, DRAWINGS THAT ARE UPDATED WITHOUT SEC PERMISSION, OR CHANGES IN FIELD CONDITIONS. ANY REPRODUCTION OR MODIFICATION WITHOUT WRITTEN PERMISSION IS STRICTLY PROHIBITED. | <div><br/>1094 Bankhead Hwy, Winder, GA 30680<br/>PH: 770-868-5599   Fax: 770-868-5735<br/>www.southeastculvert.com</div> | LIDL #1517 | ELBOW #6 |  | DESIGNED BY: | PROJECT NUMBER: |        |             |
|------|-------------|------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|--|--------------|-----------------|--------|-------------|
| #### | ####        | #### | #### |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                |            |          |  | DRAWN BY:    | SHEET NO. OF    |        |             |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                |            |          |  |              | MHM             | 8 / 12 |             |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                |            |          |  | DATE:        | SCALE:          |        | CHECKED BY: |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                |            |          |  | 4/1/26       | NA              |        |             |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                |            |          |  |              |                 |        |             |

\*14 GAGE, 5X1, AT2.

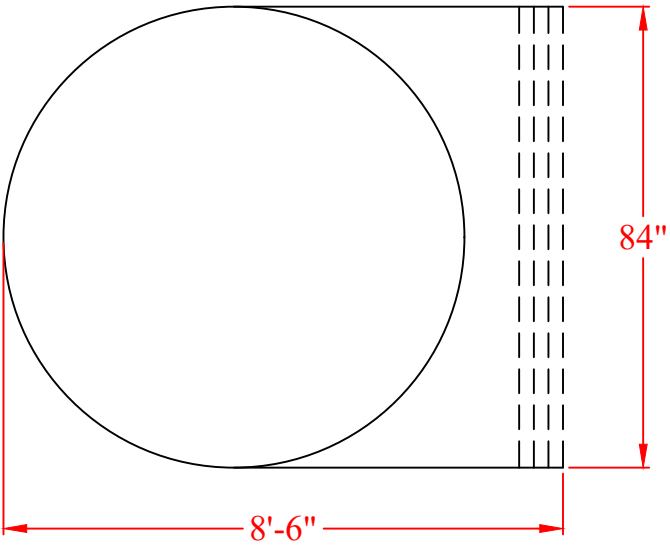


| REV  | DESCRIPTION | BY   | DATE | PROPRIETARY & CONFIDENTIAL: THIS DRAWING IS THE SOLE PROPERTY OF SOUTHEAST CULVERT, INC. ("SEC") AND CONTAINS PROPRIETARY INFORMATION APPLICABLE TO THIS PROJECT ONLY. ANY DIFFERENCES BETWEEN THE DESIGN INFORMATION, THE GENERAL NOTES AND FIELD CONDITIONS MUST BE REPORTED TO SEC IMMEDIATELY FOR REEVALUATION. SEC IS NOT LIABLE FOR INCOMPLETE OR INACCURATE DRAWINGS, DRAWINGS THAT ARE UPDATED WITHOUT SEC PERMISSION, OR CHANGES IN FIELD CONDITIONS. ANY REPRODUCTION OR MODIFICATION WITHOUT WRITTEN PERMISSION IS STRICTLY PROHIBITED. | <br>1094 Bankhead Hwy, Winder, GA 30680<br>PH: 770-868-5599   Fax: 770-868-5735<br>www.southeastculvert.com | LIDL #1517 | MANIFOLD #7 |        | DESIGNED BY: | PROJECT NUMBER: |
|------|-------------|------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|--------|--------------|-----------------|
| #### | ####        | #### | #### |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                  |            |             |        | DRAWN BY:    | SHEET NO. OF    |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                  |            |             |        | MHM          |                 |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                  |            |             |        | CHECKED BY:  | 10 / 12         |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                  |            |             |        |              |                 |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                  |            |             |        |              |                 |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                  |            | DATE:       | SCALE: |              |                 |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                  |            | 4/1/26      | NA     |              |                 |

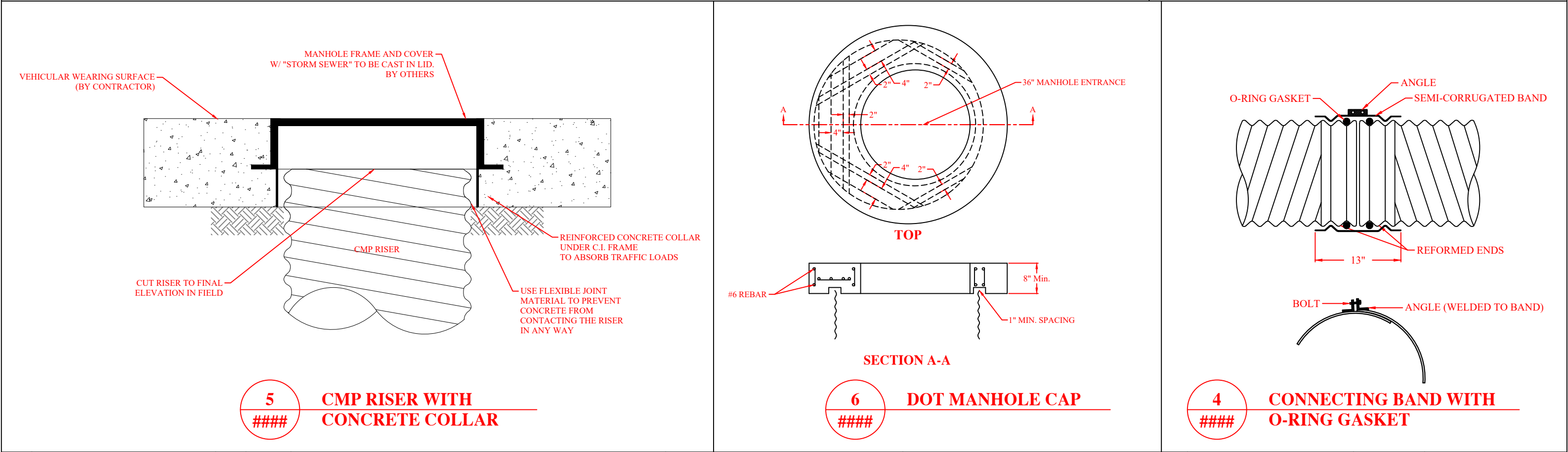
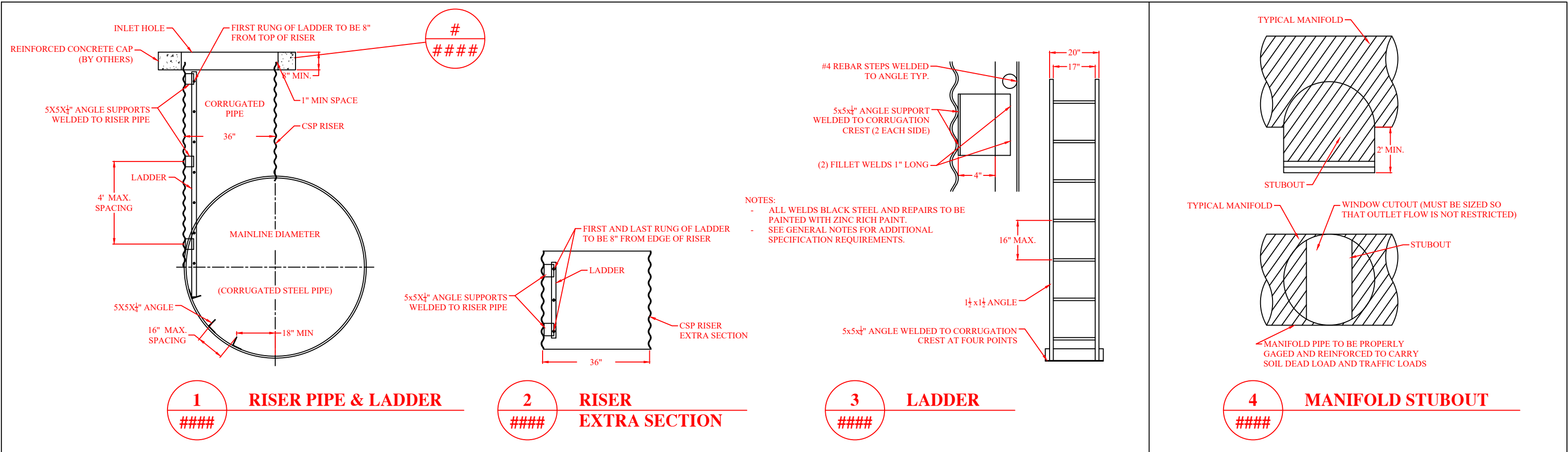
3" X 3" ANGLE REINFORCEMENT  
4.5' EACH WAY, TOP & BOTTOM



\*14 GAGE, 5X1, AT2.



|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                |            |             |        |                  |                 |
|------|-------------|------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|--------|------------------|-----------------|
| REV  | DESCRIPTION | BY   | DATE | PROPRIETARY & CONFIDENTIAL: THIS DRAWING IS THE SOLE PROPERTY OF SOUTHEAST CULVERT, INC. ("SEC") AND CONTAINS PROPRIETARY INFORMATION APPLICABLE TO THIS PROJECT ONLY. ANY DIFFERENCES BETWEEN THE DESIGN INFORMATION, THE GENERAL NOTES AND FIELD CONDITIONS MUST BE REPORTED TO SEC IMMEDIATELY FOR REEVALUATION. SEC IS NOT LIABLE FOR INCOMPLETE OR INACCURATE DRAWINGS, DRAWINGS THAT ARE UPDATED WITHOUT SEC PERMISSION, OR CHANGES IN FIELD CONDITIONS. ANY REPRODUCTION OR MODIFICATION WITHOUT WRITTEN PERMISSION IS STRICTLY PROHIBITED. | <div><br/>1094 Bankhead Hwy, Winder, GA 30680<br/>PH: 770-868-5599   Fax: 770-868-5735<br/>www.southeastculvert.com</div> | LIDL #1517 | MANIFOLD #8 |        | DESIGNED BY:     | PROJECT NUMBER: |
| #### | ####        | #### | #### |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                |            |             |        | DRAWN BY:<br>MHM | SHEET NO. OF    |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                |            |             |        | CHECKED BY:      | 10 / 12         |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                |            | DATE:       | SCALE: |                  |                 |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                |            | 4/1/26      | NA     |                  |                 |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                |            |             |        |                  |                 |



| REV  | DESCRIPTION | BY   | DATE | PROPRIETARY & CONFIDENTIAL: THIS DRAWING IS THE SOLE PROPERTY OF SOUTHEAST CULVERT, INC. ("SEC") AND CONTAINS PROPRIETARY INFORMATION APPLICABLE TO THIS PROJECT ONLY. ANY DIFFERENCES BETWEEN THE DESIGN INFORMATION, THE GENERAL NOTES AND FIELD CONDITIONS MUST BE REPORTED TO SEC IMMEDIATELY FOR REEVALUATION. SEC IS NOT LIABLE FOR INCOMPLETE OR INACCURATE DRAWINGS, DRAWINGS THAT ARE UPDATED WITHOUT SEC PERMISSION, OR CHANGES IN FIELD CONDITIONS. ANY REPRODUCTION OR MODIFICATION WITHOUT WRITTEN PERMISSION IS STRICTLY PROHIBITED. |  |  | <br>1094 Bankhead Hwy, Winder, GA 30680<br>PH: 770-868-5599   Fax: 770-868-5735<br>www.southeastculvert.com |  | LIDL #1517 |  | DETAIL 1 |        | DESIGNED BY: | PROJECT NUMBER: |
|------|-------------|------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------|--|----------|--------|--------------|-----------------|
| #### | ####        | #### | #### |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |                                                                                                                                                                                                  |  |            |  |          |        | DRAWN BY:    | SHEET NO. OF    |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |                                                                                                                                                                                                  |  |            |  |          |        | MHM          |                 |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |                                                                                                                                                                                                  |  |            |  |          | DATE:  | CHECKED BY:  | 11 / 12         |
|      |             |      |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |  |                                                                                                                                                                                                  |  |            |  |          | 4/1/26 | NA           |                 |

WELL GRADED SOILS  
COMPACTED TO 90% MIN.  
PROCTOR DENSITY

ENGINEERED FABRIC

FABRIC TO WRAP  
OUTSIDE EDGE OF  
STONE AS WELL

3/8" MIN. STONE  
3" MAX STONE  
COMPACTED TO  
90% MIN. DENSITY

ENGINEERED FABRIC

12" MIN

3' SPACING

BOTTOM OF FLEXIBLE PAVEMENT  
OR TOP OF RIGID PAVEMENT

6" 12" MIN

84" Ø CMP

6" MIN

NOTE:  
AASHTO A1 IS THE PREFERRED BACKFILL MATERIAL. HOWEVER, OTHER GRANULAR MATERIALS MAY BE USED. BECAUSE DETENTION SYSTEMS ARE OFTEN EXPOSED TO HEAVY CONSTRUCTION LOADS DURING TOTAL COMPLETION OF THE PROJECT, THERE IS NEED FOR EXTRA CARE.  
TO HANDLE THESE LOADS, IF THE MINIMUM COVER IS MORE THAN THE DIAMETER/6 AND THE DIAMETER IS LESS THAN 84", A2 OR A3 MATERIAL MAY GENERALLY BE USED IF THEY ARE PLACED IN 8" LIFTS AND COMPACTED TO 90% PROCTOR DENSITY. AS PIPE DIAMETERS INCREASE AND COVER DECREASES(APPROACHING THE MINIMUM COVER LEVEL), THE QUALITY OF THE BACKFILL AND COMPACTION DENSITY BECOME MORE IMPORTANT.

8

DETENTION BACKFILL

####

CONSTRUCTION EQUIPMENT TRAFFIC

FINISHED GRADE

CONSTRUCTION COVER

TEMPORARY COVER FOR CONSTRUCTION LOADS

| Minimum Cover for Construction Loads |            |            |             |              |
|--------------------------------------|------------|------------|-------------|--------------|
| Pipe Span, in.                       | 18-50 kips | 50-75 kips | 75-110 kips | 110-150 kips |
| 12-42                                | 2.0 ft     | 2.5 ft     | 3.0 ft      | 3.0 ft       |
| 48-72                                | 3.0 ft     | 3.0 ft     | 3.5 ft      | 4.0 ft       |
| 78-120                               | 3.0 ft     | 3.5 ft     | 4.0 ft      | 4.0 ft       |
| 126-144                              | 3.5 ft     | 4.0 ft     | 4.5 ft      | 4.5 ft       |

9

CONSTRUCTION COVER

####

MAINLINE

CROSS-SECTION

MAINLINE DIAMETER  
(CORRUGATED STEEL PIPE)

BAND TAPERED SLEEVE  
TO CSP STUBOUT

RAM TAPERED SLEEVE  
TO TIGHT FIT

STEEL TAPERED SLEEVE

CSP STUBOUT

HDPE PIPE  
(OR RCP)

X" X" X-0.25" X"

11

TAPERED SLEEVE

####

| REV  | DESCRIPTION | BY   | DATE |
|------|-------------|------|------|
| #### | ####        | #### | #### |
|      |             |      |      |
|      |             |      |      |
|      |             |      |      |
|      |             |      |      |

PROPRIETARY & CONFIDENTIAL: THIS DRAWING IS THE SOLE PROPERTY OF SOUTHEAST CULVERT, INC. ("SEC") AND CONTAINS PROPRIETARY INFORMATION APPLICABLE TO THIS PROJECT ONLY. ANY DIFFERENCES BETWEEN THE DESIGN INFORMATION, THE GENERAL NOTES AND FIELD CONDITIONS MUST BE REPORTED TO SEC IMMEDIATELY FOR REEVALUATION. SEC IS NOT LIABLE FOR INCOMPLETE OR INACCURATE DRAWINGS, DRAWINGS THAT ARE UPDATED WITHOUT SEC PERMISSION, OR CHANGES IN FIELD CONDITIONS. ANY REPRODUCTION OR MODIFICATION WITHOUT WRITTEN PERMISSION IS STRICTLY PROHIBITED.

S

e

C

SOUTHEAST  
CULVERT

1094 Bankhead Hwy, Winder, GA 30680  
PH: 770-868-5599 | Fax: 770-868-5735  
www.southeastculvert.com

LIDL #1517

DETAIL 2

DATE: 4/1/26SCALE: NA

DESIGNED BY:

DRAWN BY: MHM

CHECKED BY:

PROJECT NUMBER:

SHEET NO. OF

12 / 12